Analysis of Functions

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July 27, 2026

Analysis of functions is the study of *spaces* of functions, and of the operations that act on them. The change of viewpoint is a small one to state and an enormous one to work with: instead of asking what a single function does at a single point, we regard a function as one point in an infinite-dimensional space, and we ask about the geometry and topology of that space.

Once we do this, a great deal becomes possible. Questions about partial differential equations turn into questions about linear operators between Banach spaces. Questions about convergence of Fourier series turn into questions about orthonormal bases in a Hilbert space. And the classical obstruction that non-differentiable functions cannot be differentiated simply dissolves: we enlarge the notion of a function until every distribution has derivatives of every order, and then we work out afterwards which of our generalised objects were honest functions all along.

The unifying technique of the course is *compactness by weakening*. In infinite dimensions the closed unit ball is never compact in the norm topology, so bounded sequences need not have convergent subsequences and the classical method of extracting limits fails. The remedy is to weaken the topology until compactness returns, at the price that the limits we obtain are much worse behaved. Recovering the good behaviour afterwards – through convexity, lower semicontinuity, elliptic regularity – is the real work, and the pay-off is the existence of solutions to nonlinear partial differential equations that no explicit formula will ever reach.

This article constitutes my notes for the 'Analysis of Functions' course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

Contents

§1 Introduction

Let me try to say in slightly more detail what the plan is.

We begin, in Section 2, with a review of measure and integration. Almost none of this is new – it is the Probability and Measure course, with the probabilistic language stripped out – but we do need it in a hurry and in a slightly different key, so it is worth setting down. The one genuinely new item is Littlewood's three principles, which are the informal slogans that make measure theory feel like analysis rather than bookkeeping.

Section 3 builds the spaces L^p. These are our first examples of Banach spaces made of functions, and essentially everything later in the course is a statement about them or about their relatives. We prove Hölder's and Minkowski's inequalities, completeness, and – most importantly for what follows – a collection of density theorems, culminating in the fact that any L^p function ($p < \infty$) can be approximated in norm by smooth compactly supported functions. The tool that does this is *mollification*: convolution with a shrinking bump. It is worth understanding it thoroughly, since it is used on almost every subsequent page.

Section 5 is about duality. We compute the dual of L^p, prove the Radon–Nikodým theorem (a measure-theoretic statement, via a Hilbert space argument – an early sign of how the subject works), and state the Riesz representation theorem which identifies measures as functionals on continuous functions. Section 6 then introduces the weak and weak-$*$ topologies, proves Hahn–Banach and Banach–Alaoglu, and so gets us the compactness we were after.

With compactness in hand, Section 7 sets out the direct method of the calculus of variations in the abstract: a coercive, lower semicontinuous functional on a suitable space attains its infimum. The delicate point is that coercivity wants a weak topology and lower semicontinuity wants a strong one, and *convexity* is what reconciles them.

Sections **??** and **??** develop the Fourier transform and distribution theory in parallel, since each is much improved by the other. The Fourier transform is an isometry (up to constants) of L^2, an isomorphism of the Schwartz space, and – once we have tempered distributions – a bijection of a class of objects large enough to include every polynomial, every bounded function, and the Dirac delta.

Finally Sections **??** and **??** put everything to work. Sobolev spaces are the function spaces adapted to differential operators; the Sobolev embedding theorem says that enough weak differentiability forces classical differentiability, and the Rellich–Kondrachov theorem provides the compactness that lets the direct method run. We finish by solving the Dirichlet problem, describing the eigenvalues of the Laplacian on a bounded domain, and proving existence of minimisers for a class of variational problems whose Euler–Lagrange equations we could not hope to solve by hand.

§2 Measure and Integration

We assume the reader has met the Lebesgue integral. This section is a review: it fixes notation, records the results we shall use constantly, and adds the three theorems of Littlewood that describe how far measurable objects are from classical ones.

§2.1 Measure spaces

We recall the basic framework.

Definition 2.1 (σ-algebra)
Let E be a set. A collection $\mathcal{E}$ of subsets of E is a **σ-algebra** on E if

(i) $\emptyset \in \mathcal{E}$;

(ii) $A \in \mathcal{E} \implies A^c = E \setminus A \in \mathcal{E}$;

(iii) if $(A_n)_{n\in\mathbb{N}}$ is a sequence in $\mathcal{E}$ then $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{E}$.

The pair $(E, \mathcal{E})$ is a **measurable space**, and elements of $\mathcal{E}$ are **measurable sets**.

Definition 2.2 (Measure)
A **measure** on a measurable space $(E, \mathcal{E})$ is a function $\mu : \mathcal{E} \to [0, \infty]$ with $\mu(\emptyset) = 0$ which is **countably additive**: if $(A_n)_{n\in\mathbb{N}}$ are disjoint elements of $\mathcal{E}$ then

$$\mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} \mu(A_n).$$

We call $(E, \mathcal{E}, \mu)$ a **measure space**.

Countable additivity is the whole content of the definition; everything else about measures is a consequence of it, together with the elementary manipulations of set theory. In particular a measure is monotone, countably subadditive, and continuous along increasing and (given finiteness) decreasing sequences of sets.

Example 2.3

On any set E we may take $\mathcal{E} = 2^E$ and $\mu(A) = \mathrm{card}(A) \in [0, \infty]$, the counting measure. This is not a frivolous example: with $E = \mathbb{N}$ the L^p spaces of the counting measure are the sequence spaces ℓ^p, and every theorem we prove about L^p spaces has a sequence space avatar which is usually easier to think about.

Definition 2.4 (Generated σ-algebra)

Given a collection $\mathcal{A}$ of subsets of E, the σ-algebra **generated** by $\mathcal{A}$ is

$$\sigma(\mathcal{A}) = \bigcap_{\substack{\mathcal{E} \supseteq \mathcal{A} \\ \mathcal{E} \text{ a } \sigma\text{-algebra}}} \mathcal{E},$$

the smallest σ-algebra containing $\mathcal{A}$. If $(E, \mathcal{T})$ is a topological space then $\mathcal{B}(E) = \sigma(\mathcal{T})$ is the **Borel σ-algebra**.

The example we care about above all others is Lebesgue measure. Its existence was the main construction of the measure theory course, and we simply quote it.

Theorem 2.5 (Lebesgue measure)

There is a σ-algebra $\mathcal{M}$ of subsets of $\mathbb{R}^n$ and a measure λ on $(\mathbb{R}^n, \mathcal{M})$ such that

(i) $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}$;

(ii) if $A = (a_1, b_1] \times \cdots \times (a_n, b_n]$ is a box, with $-\infty < a_i \le b_i < \infty$, then

$$\lambda(A) = (b_1 - a_1) \cdots (b_n - a_n);$$

(iii) (**Borel regularity**) $A \in \mathcal{M}$ if and only if for every $\varepsilon > 0$ there are a closed set C and an open set O with $C \subseteq A \subseteq O$ and $\lambda(O \setminus C) < \varepsilon$.

These conditions determine $\mathcal{M}$ and λ uniquely.

We write dx for $d\lambda$ and $|A|$ for $\lambda(A)$ when no confusion can arise. Property (iii) deserves emphasis, because it is the one we shall reach for repeatedly and it is easy to forget that it is available. It says that a Lebesgue measurable set is, up to arbitrarily small error, both an open set and a closed set. This is the rigorous form of Littlewood's first principle, and it is the engine behind almost every approximation argument in the course: if we want to prove something for all measurable sets, it is usually enough to prove it for boxes and then turn the handle.

Definition 2.6 (Measurable function)
Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measurable spaces. A function $f : E \to G$ is **measurable** if $f^{-1}(A) \in \mathcal{E}$ for all $A \in \mathcal{G}$.

When $(G, \mathcal{G}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ or $(\mathbb{C}, \mathcal{B}(\mathbb{C}))$ we just say 'f is measurable on E'; when $(G, \mathcal{G}) = ([0, \infty], \mathcal{B}([0, \infty]))$ we say f is a non-negative measurable function. Recall that measurability is preserved by the vector space operations, by products, by countable suprema and infima, and by pointwise (indeed almost everywhere) limits. This last stability property is exactly what the Riemann theory lacks and is the reason we work with the Lebesgue integral at all.

§2.2 The integral

Definition 2.7 (Simple function)
A **simple function** on $(E, \mathcal{E})$ is a function of the form $f = \sum_{k=1}^{N} a_k \mathbb{1}_{A_k}$ with a_k scalars and $A_k \in \mathcal{E}$.

For a non-negative simple function (which we may always write with $a_k \in [0, \infty]$ and A_k disjoint) we define

$$\mu(f) = \int_E f \, \mathrm{d}\mu = \sum_{k=1}^{N} a_k \mu(A_k),$$

with the convention $0 \cdot \infty = \infty \cdot 0 = 0$. For general non-negative measurable f we set

$$\mu(f) = \int_E f \, \mathrm{d}\mu = \sup\{\mu(g) : 0 \leq g \leq f, \ g \text{ simple}\},$$

and finally a measurable f is **integrable** if $\mu(|f|) < \infty$, in which case writing $f = f^+ - f^-$ with $f^\pm \geq 0$ we set $\mu(f) = \mu(f^+) - \mu(f^-)$. Complex-valued f are handled by taking real and imaginary parts.

The integral so defined is linear, monotone, and additive over disjoint sets. The two theorems that make it usable are the following.

Theorem 2.8 (Monotone convergence)
If f_n are non-negative measurable with $f_n \uparrow f$ almost everywhere, then $\int_E f_n \, \mathrm{d}\mu \uparrow \int_E f \, \mathrm{d}\mu$.

Theorem 2.9 (Dominated convergence)
Suppose $f_n \to f$ almost everywhere and $|f_n| \leq g$ almost everywhere for some integrable g. Then f is integrable and $\int_E f_n \, \mathrm{d}\mu \to \int_E f \, \mathrm{d}\mu$.

We shall also use Fatou's lemma ($\int \liminf f_n \leq \liminf \int f_n$ for non-negative f_n) and Fubini–Tonelli, which lets us exchange the order of a double integral provided either the integrand is non-negative or the double integral converges absolutely. These will be used without comment.

Remark. A remark on the meaning of '$f \in L^p$' that is worth making early. Two functions agreeing almost everywhere have the same integral against everything, and there is no measure-theoretic way to tell them apart. We therefore work throughout with equivalence classes of functions modulo equality almost everywhere, while continuing to write $f(x)$ as though f were a function. This costs us nothing until we want to restrict f to a set of measure zero – for instance to a hypersurface, when solving a boundary value problem – and then it costs us a whole theorem (the trace theorem, Section **??**).

§2.3 Littlewood's three principles

Littlewood wrote in 1944 that there are three principles, roughly expressible as follows: *every measurable set is nearly a finite union of intervals; every measurable function is nearly continuous; every convergent sequence of measurable functions is nearly uniformly convergent.* The word 'nearly' means 'up to a set of arbitrarily small measure'. Each principle is a theorem, and it is instructive to see them proved, both because we use them and because their proofs are a compendium of the standard tricks.

The first principle is Borel regularity, essentially by definition of Lebesgue measure.

Proposition 2.10

Let $A \subseteq \mathbb{R}^n$ be measurable with $|A| < \infty$. Then for every $\varepsilon > 0$ there is a finite union B of boxes with $|A \triangle B| < \varepsilon$.

Proof. By Theorem 2.5(iii) choose an open $O \supseteq A$ with $|O \setminus A| < \varepsilon/2$. Every open subset of $\mathbb{R}^n$ is a countable disjoint union of half-open dyadic boxes $\bigcup_k Q_k$; since $|O| \le |A| + \varepsilon/2 < \infty$, the tail $\sum_{k>N} |Q_k|$ can be made smaller than $\varepsilon/2$. Taking $B = \bigcup_{k \le N} Q_k$ gives

$$|A \triangle B| \le |O \setminus A| + |O \setminus B| < \varepsilon.$$

□

The third principle is Egorov's theorem. Note that some hypothesis of finite measure is necessary: on $\mathbb{R}$ the functions $f_n = \mathbb{1}_{[n,n+1]}$ converge pointwise to 0 but not uniformly off any set of finite measure.

Theorem 2.11 (Egorov)

Let $E \subseteq \mathbb{R}^d$ be measurable with $|E| < \infty$, and let $f_n, f : E \to \mathbb{C}$ be measurable with $f_n \to f$ almost everywhere on E. Then for every $\varepsilon > 0$ there is a closed set $A_\varepsilon \subseteq E$ with $|E \setminus A_\varepsilon| < \varepsilon$ and $f_n \to f$ uniformly on A_ε.

Proof. Discarding a null set, we may assume $f_n \to f$ everywhere on E. For $n, k \in \mathbb{N}$ put

$$E_k^n = \{x \in E : |f_j(x) - f(x)| < 1/n \text{ for all } j > k\}.$$

For fixed n these increase in k, and their union over k is all of E by pointwise convergence. Since $|E| < \infty$ we have $|E_k^n| \uparrow |E|$, so we may choose k_n with $|E \setminus E_{k_n}^n| < 2^{-n}\varepsilon$.

Set $A' = \bigcap_{n \ge 1} E_{k_n}^n$. Then

$$|E \setminus A'| \le \sum_{n \ge 1} |E \setminus E_{k_n}^n| < \varepsilon,$$

after replacing ε by $\varepsilon/2$ if we like. And the convergence is uniform on A': given $\delta > 0$ pick $n > 1/\delta$; then for every $x \in A' \subseteq E^n_{k_n}$ and every $j > k_n$ we have $|f_j(x) - f(x)| < 1/n < \delta$, and the choice of k_n did not depend on x. Finally, Borel regularity lets us shrink A' to a closed A_ε with $|A' \setminus A_\varepsilon|$ as small as we please. $\square$

The second principle is Lusin's theorem, and its proof uses the third.

Theorem 2.12 (Lusin)
Let $E \subseteq \mathbb{R}^d$ be measurable with $|E| < \infty$ and let f be measurable and almost everywhere finite on E. Then for every $\varepsilon > 0$ there is a closed $F_\varepsilon \subseteq E$ with $|E \setminus F_\varepsilon| < \varepsilon$ such that $f|_{F_\varepsilon}$ is continuous.

Proof. Suppose first that $f = \sum_{n=1}^m a_n \mathbb{1}_{A_n}$ is simple, with the A_n a partition of E. By Borel regularity choose compact $K_n \subseteq A_n$ with $|A_n \setminus K_n| < \varepsilon/m$, and set $B = \bigcup_{n=1}^m K_n$, so that $|E \setminus B| < \varepsilon$. The sets K_n are compact and disjoint, hence pairwise at positive distance; as f is constant on each K_n it is continuous on B.

For general f, choose simple $f_n \to f$ almost everywhere, and by the previous paragraph choose measurable C_n with $|C_n| < 2^{-n}$ and f_n continuous on $E \setminus C_n$. By Egorov there is $A \subseteq E$ with $|E \setminus A| < \varepsilon/3$ and $f_n \to f$ uniformly on A. Pick N with $\sum_{n \geq N} 2^{-n} < \varepsilon/3$ and set

$$F' = A \setminus \bigcup_{n \geq N} C_n,$$

so $|E \setminus F'| < 2\varepsilon/3$. On F' each f_n $(n \geq N)$ is continuous and $f_n \to f$ uniformly, so $f|_{F'}$ is continuous. Borel regularity again produces a closed $F_\varepsilon \subseteq F'$ of nearly the same measure. $\square$

Remark. A word of warning, because this trips up everyone at least once. To say '$f|_F$ is continuous' is a statement about f restricted to F with its subspace topology; it is emphatically *not* the statement that f, as a function on E, is continuous at each point of F. For a spectacular example take $f = \mathbb{1}_{\mathbb{Q}}$ on $\mathbb{R}$. Then f is continuous at no point whatever, but $f|_{\mathbb{R}\setminus\mathbb{Q}} \equiv 0$ is as continuous as one could wish.

§3 L^p Spaces

Every measure space carries with it a family of Banach spaces, one for each $p \in [1, \infty]$, and these are the spaces in which we shall spend the rest of the course. Their theory is a rare thing in analysis: it is genuinely uniform in p, so that a single argument handles a continuum of spaces, and yet the endpoints $p = 1$ and $p = \infty$ are always exceptional, and the exceptions are always interesting.

§3.1 Definitions

Definition 3.1 (L^p norm)
Let $(E, \mathcal{E}, \mu)$ be a measure space and $f : E \to \mathbb{C}$ measurable. For $1 \leq p < \infty$ we

define

$$\|f\|_{L^p} = \left(\int_E |f|^p \, \mathrm{d}\mu\right)^{1/p},$$

and we define the **essential supremum**

$$\|f\|_{L^\infty} = \operatorname*{ess\,sup}_E |f| = \inf\{K \geq 0 : |f| \leq K \text{ almost everywhere}\}.$$

For $1 \leq p \leq \infty$, the **L^p space** over E is

$$L^p(E, \mu) = \{f : E \to \mathbb{C} \text{ measurable} : \|f\|_{L^p} < \infty\}/\sim,$$

where $f \sim g$ if $f = g$ almost everywhere.

When $(E, \mathcal{E}, \mu) = (\mathbb{R}^n, \mathcal{M}, \lambda)$ we write simply $L^p(\mathbb{R}^n)$, and when $\Omega \subseteq \mathbb{R}^n$ is measurable, $L^p(\Omega)$ means $L^p(\Omega, \lambda|_\Omega)$. The quotient by almost-everywhere equality is not a technicality we can dispense with: without it $\|\cdot\|_{L^p}$ is only a seminorm, since a function supported on a null set has norm zero.

Note that the infimum in the definition of ess sup is attained – if $|f| \leq K_n$ a.e. and $K_n \downarrow K$ then $|f| \leq K$ off the union of countably many null sets – so $|f| \leq \|f\|_{L^\infty}$ almost everywhere, as one would hope.

Example 3.2

Let $E = \mathbb{N}$ with counting measure. Then $L^p(\mathbb{N})$ is the sequence space ℓ^p of sequences (a_n) with $\sum |a_n|^p < \infty$ (with the obvious modification at $p = \infty$). Since no non-empty set has measure zero, there is no quotient to take, and 'almost everywhere' means 'everywhere'.

Example 3.3

Let $E = [0, 1]$ with Lebesgue measure and $f(x) = x^{-\alpha}$ for $\alpha > 0$. Then $\int_0^1 x^{-\alpha p} \, \mathrm{d}x < \infty$ if and only if $\alpha p < 1$, so $f \in L^p([0, 1])$ exactly when $p < 1/\alpha$. Thus by varying α we can produce a function lying in L^p for all p below any prescribed threshold and in no L^p above it. The spaces really are different.

§3.2 Hölder and Minkowski

The two fundamental inequalities both come from convexity. We say $p, q \in [1, \infty]$ are **conjugate exponents** if

$$\frac{1}{p} + \frac{1}{q} = 1,$$

with the convention $1/\infty = 0$; so 1 and ∞ are conjugate, and 2 is self-conjugate.

Lemma 3.4 (Young's inequality)

Let $p, q \in (1, \infty)$ be conjugate and $a, b \geq 0$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality if and only if $a^p = b^q$.

Proof. If a or b is zero this is clear, so assume both positive. The logarithm is strictly concave, so

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p}\log(a^p) + \frac{1}{q}\log(b^q) = \log a + \log b = \log(ab),$$

using $1/p + 1/q = 1$, and strict concavity gives equality precisely when $a^p = b^q$. Exponentiating gives the result. □

Theorem 3.5 (Hölder's inequality)
Let $p, q \in [1, \infty]$ be conjugate and let $f \in L^p(E, \mu)$, $g \in L^q(E, \mu)$. Then $fg \in L^1(E, \mu)$ and

$$\|fg\|_{L^1} \leq \|f\|_{L^p} \|g\|_{L^q}.$$

Proof. The cases $p = 1, q = \infty$ (and its mirror) are immediate from $|fg| \leq |f| \|g\|_{L^\infty}$ almost everywhere. So suppose $1 < p < \infty$. If either norm is zero the corresponding function vanishes a.e. and there is nothing to prove; and if either is infinite there is nothing to prove either. So normalise: let $F = |f| / \|f\|_{L^p}$ and $G = |g| / \|g\|_{L^q}$, so that $\|F\|_{L^p} = \|G\|_{L^q} = 1$. By Lemma 3.4 applied pointwise,

$$FG \leq \frac{F^p}{p} + \frac{G^q}{q},$$

and integrating,

$$\int_E FG \, \mathrm{d}\mu \leq \frac{1}{p}\int_E F^p \, \mathrm{d}\mu + \frac{1}{q}\int_E G^q \, \mathrm{d}\mu = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying through by $\|f\|_{L^p} \|g\|_{L^q}$ gives the claim. □

Hölder's inequality is the reason conjugate exponents are worth naming. It says the pairing $(f, g) \mapsto \int fg$ is bounded on $L^p \times L^q$, and we shall see in Section 5 that it accounts for *all* bounded linear functionals on L^p when $p < \infty$.

Theorem 3.6 (Minkowski's inequality)
Let $1 \leq p \leq \infty$ and $f, g \in L^p(E, \mu)$. Then $f + g \in L^p(E, \mu)$ and

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

Proof. The cases $p = 1$ and $p = \infty$ are immediate from the pointwise triangle inequality, so take $1 < p < \infty$. Since $|f + g|^p \leq (2\max(|f|, |g|))^p \leq 2^p(|f|^p + |g|^p)$, the left-hand side is finite. Now write

$$\int_E |f + g|^p \, \mathrm{d}\mu \leq \int_E |f| \, |f + g|^{p-1} \, \mathrm{d}\mu + \int_E |g| \, |f + g|^{p-1} \, \mathrm{d}\mu,$$

and apply Hölder to each term, noting that $|f+g|^{p-1} \in L^q$ with

$$\left\| |f+g|^{p-1} \right\|_{L^q} = \left(\int_E |f+g|^{(p-1)q} \, d\mu \right)^{1/q} = \|f+g\|_{L^p}^{p/q}$$

since $(p-1)q = p$. This gives

$$\|f+g\|_{L^p}^p \le (\|f\|_{L^p} + \|g\|_{L^p}) \, \|f+g\|_{L^p}^{p/q}.$$

If $\|f+g\|_{L^p} = 0$ we are done; otherwise divide by $\|f+g\|_{L^p}^{p/q}$ and use $p - p/q = 1$. $\square$

So $L^p(E,\mu)$ is a normed vector space. We will also want the following integral form of Minkowski's inequality, which says that 'the norm of an integral is at most the integral of the norms' – a continuous version of the triangle inequality. Its proof is the same duality argument as above and is left as an exercise.

Lemma 3.7 (Minkowski's integral inequality)
Let $F : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}$ be measurable with $F(\cdot, y) \in L^p$ for a.e. y, and let $1 \le p < \infty$. Then

$$\left(\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x,y) \, dy \right|^p dx \right)^{1/p} \le \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x,y)|^p \, dx \right)^{1/p} dy.$$

§3.3 Completeness

The next theorem is the reason L^p spaces are useful rather than merely definable.

Theorem 3.8 (Riesz–Fischer)
For every $1 \le p \le \infty$, the space $(L^p(E,\mu), \|\cdot\|_{L^p})$ is a Banach space.

Proof. Recall that a normed space is complete if and only if every absolutely convergent series converges. So let (f_k) be a sequence in L^p with $\sum_k \|f_k\|_{L^p} = M < \infty$.

Take first $p < \infty$. Set $g_N = \sum_{k=1}^N |f_k|$ and $g = \sum_{k=1}^\infty |f_k| \in [0,\infty]$, so $g_N \uparrow g$ pointwise. By Minkowski $\|g_N\|_{L^p} \le M$ for every N, and by monotone convergence applied to $g_N^p \uparrow g^p$,

$$\int_E g^p \, d\mu = \lim_N \int_E g_N^p \, d\mu \le M^p.$$

In particular $g < \infty$ almost everywhere, so the series $\sum_k f_k(x)$ converges absolutely for almost every x; call its sum $f(x)$ (and set $f = 0$ on the null set where it does not converge). Then $|f| \le g$, so $f \in L^p$. Moreover

$$\left| f - \sum_{k=1}^N f_k \right|^p \le \left(\sum_{k>N} |f_k| \right)^p \le g^p \in L^1,$$

and the left-hand side tends to 0 pointwise a.e., so dominated convergence gives $\left\| f - \sum_{k \le N} f_k \right\|_{L^p} \to 0$ as required.

For $p = \infty$, let $A_k = \{|f_k| > \|f_k\|_{L^\infty}\}$, a null set, and let $A = \bigcup_k A_k$, also null. Off A the series $\sum_k f_k$ converges uniformly by the Weierstrass M-test, to a bounded limit f, and uniform convergence off a null set is exactly convergence in L^∞. □

The $p = 2$ case is special, because the norm comes from an inner product

$$\langle f, g\rangle_{L^2} = \int_E \bar{f} g \, \mathrm{d}\mu,$$

which is finite by Hölder (this is Cauchy–Schwarz). So $L^2(E, \mu)$ is a Hilbert space, and we may use all the geometry – orthogonal projection, orthonormal bases, the Riesz representation theorem – that Hilbert space affords. Much of what we do later works only for $p = 2$ for exactly this reason.

§3.4 Inclusions between L^p spaces

It is natural to ask how the spaces $L^p(E, \mu)$ sit relative to one another as p varies. The answer depends entirely on the measure space, and the two extreme cases are worth having clearly in mind.

Proposition 3.9

Suppose $\mu(E) < \infty$ and $1 \le p \le q \le \infty$. Then $L^q(E, \mu) \subseteq L^p(E, \mu)$, and

$$\|f\|_{L^p} \le \mu(E)^{\frac{1}{p} - \frac{1}{q}} \|f\|_{L^q}.$$

Proof. If $q = \infty$ then $\int |f|^p \le \|f\|_{L^\infty}^p \mu(E)$ directly. Otherwise apply Hölder to the pair $|f|^p \in L^{q/p}$ and $1 \in L^r$ where r is conjugate to q/p:

$$\int_E |f|^p \, \mathrm{d}\mu \le \left(\int_E |f|^q \, \mathrm{d}\mu \right)^{p/q} \mu(E)^{1 - p/q},$$

and take p-th roots. □

On a space of infinite measure the situation is completely different: there are *no* inclusions at all between distinct $L^p(\mathbb{R}^n)$. A function can fail to be in L^p for two quite different reasons – it can be too singular locally, or it can decay too slowly at infinity – and raising p helps with one while hurting the other.

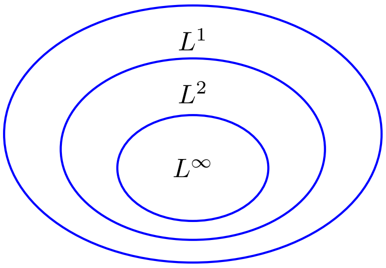
Example 3.10

Work on $\mathbb{R}$, and let $1 \le p < q < \infty$. Then

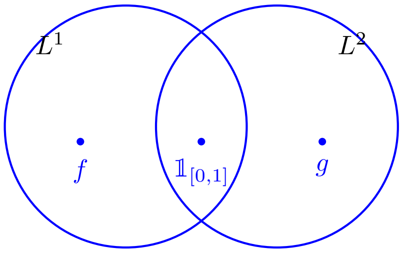
$$f(x) = |x|^{-1/q} \mathbb{1}_{\{|x|<1\}} \in L^p \setminus L^q, \qquad g(x) = |x|^{-1/p} \mathbb{1}_{\{|x|>1\}} \in L^q \setminus L^p.$$

Indeed $\int_0^1 x^{-p/q} \, \mathrm{d}x < \infty$ since $p/q < 1$, while $\int_0^1 x^{-1} \, \mathrm{d}x = \infty$; and $\int_1^\infty x^{-q/p} \, \mathrm{d}x < \infty$ since $q/p > 1$, while $\int_1^\infty x^{-1} \, \mathrm{d}x = \infty$.

The following picture summarises the two situations. On the left, a finite measure space, where the L^p spaces are nested and higher exponents mean smaller spaces; on the right, $\mathbb{R}^n$, where the spaces genuinely overlap and neither contains the other.



$\mu(E) < \infty$



$E = \mathbb{R}^n$

There is one general statement that survives on any measure space, and it is a useful one: the L^p norms are logarithmically convex in $1/p$.

Proposition 3.11 (Interpolation)
Let $1 \leq p < r < q \leq \infty$ and write $\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}$ for some $\theta \in (0,1)$. Then for measurable f,

$$\|f\|_{L^r} \leq \|f\|_{L^p}^{\theta} \, \|f\|_{L^q}^{1-\theta}.$$

In particular $L^p \cap L^q \subseteq L^r$.

Proof. Suppose $q < \infty$; the case $q = \infty$ is an easier variant. Write $|f|^r = |f|^{\theta r}|f|^{(1-\theta)r}$ and apply Hölder with the conjugate pair $\frac{p}{\theta r}$ and $\frac{q}{(1-\theta)r}$, which are indeed conjugate by the definition of θ:

$$\int |f|^r \leq \left(\int |f|^p\right)^{\theta r/p} \left(\int |f|^q\right)^{(1-\theta)r/q}.$$

Taking r-th roots gives the result. □

§3.5 Density of simple functions

Every approximation theorem in this course begins with simple functions, so let us record the basic statement.

Proposition 3.12
Let $1 \leq p < \infty$. The set of simple functions s with $\mu(\{s \neq 0\}) < \infty$ is dense in $L^p(E,\mu)$. For $p = \infty$, the simple functions are dense, but the finite-support condition cannot be imposed.

Proof. Let $f \in L^p$ and, splitting into real and imaginary and then positive and negative parts, assume $f \geq 0$. Choose simple $0 \leq s_n \uparrow f$ pointwise (the standard dyadic approximation does this). Then $|f - s_n|^p \leq f^p \in L^1$ and $|f - s_n|^p \to 0$ pointwise, so $\|f - s_n\|_{L^p} \to 0$ by dominated convergence. Each s_n has $\mu(\{s_n \neq 0\}) < \infty$: indeed $s_n \leq f$, so on $\{s_n \geq \varepsilon\}$ we have $\mu(\{s_n \geq \varepsilon\}) \leq \varepsilon^{-p} \|f\|_{L^p}^p < \infty$ by Chebyshev, and s_n takes finitely many values.

For $p = \infty$ the dyadic approximation converges uniformly on the set where f is bounded, which is almost everything; but a simple function of finite support cannot approximate the constant function 1 on $\mathbb{R}^n$ in L^∞. □

Simple functions are, as their name suggests, simple, but they are not *pleasant*: they are discontinuous, and one cannot differentiate them. In Section 4 we shall upgrade this density statement dramatically, to smooth compactly supported functions. That is a much more useful statement, and the mechanism which produces it – mollification – is one of the central tools of the course.

§3.6 Separability

Definition 3.13

A metric space is **separable** if it has a countable dense subset.

Separability is a smallness condition, and we shall need it in two places: to run a diagonal argument in the proof of Banach–Alaoglu, and to know that L^2 has a countable orthonormal basis.

Proposition 3.14

For $1 \le p < \infty$, the space $L^p(\mathbb{R}^n)$ is separable. The space $L^\infty(\mathbb{R}^n)$ is not.

Proof. For separability, consider the countable family $\mathcal{C}$ of finite rational linear combinations of indicators of boxes with rational vertices. Given $f \in L^p$ and $\varepsilon > 0$, Proposition 3.12 produces a simple $s = \sum_{k \le N} a_k \mathbb{1}_{A_k}$ with finite-measure A_k and $\|f - s\|_{L^p} < \varepsilon/2$. Each $\mathbb{1}_{A_k}$ is approximated in L^p by an indicator of a finite union of rational boxes, by Proposition 2.10 together with $\|\mathbb{1}_A - \mathbb{1}_B\|_{L^p} = |A \triangle B|^{1/p}$; and each a_k may be replaced by a nearby rational. So $\mathcal{C}$ is dense.

For L^∞, note that $\left\|\mathbb{1}_{B_r(0)} - \mathbb{1}_{B_s(0)}\right\|_{L^\infty} = 1$ whenever $r \ne s$. This is an uncountable family of points pairwise at distance 1, and no countable set can be dense in such a space (each point of the family needs its own approximant). □

§4 Convolution and Mollification

We now come to the single most useful technique in the course. The idea is very simple. Averaging a function against a small smooth bump produces a smooth function, and if the bump is small enough the average is close to the original. So every L^p function ($p < \infty$) is a limit of smooth functions, and any statement which is stable under L^p limits need only be checked for smooth functions.

§4.1 Convolution

Definition 4.1 (Convolution)

For measurable $f, g : \mathbb{R}^n \to \mathbb{C}$ we define the **convolution**

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y) g(x - y) \, \mathrm{d}y,$$

whenever the integral exists.

The integral certainly exists for every x when, say, $f \in L^1(\mathbb{R}^n)$ and $g \in L^\infty(\mathbb{R}^n)$, or when $f, g \in C_c(\mathbb{R}^n)$. More generally, Hölder shows the integral converges absolutely when $f \in L^p$ and $g \in L^q$ for conjugate p, q. The following bookkeeping lemma records the algebraic properties.

Lemma 4.2

For $f, g, h \in C_c^\infty(\mathbb{R}^n)$ we have $f * g = g * f$ and $(f * g) * h = f * (g * h)$. Moreover

$$\int_{\mathbb{R}^n} f * g = \left(\int_{\mathbb{R}^n} f\right)\left(\int_{\mathbb{R}^n} g\right).$$

Proof. Commutativity is the substitution $z = x - y$; associativity and the last identity are Fubini together with a change of variables, all of which is justified since everything in sight is continuous and compactly supported. (The hypotheses can be weakened considerably – for instance the last identity holds for $f, g \in L^1$ – but this version suffices for us.) $\square$

Convolution is very much a smoothing operation, in a sense the next result makes precise: $f * g$ inherits the *best* regularity of its two factors, not the worst. To state it we need the standard notation for higher derivatives.

Definition 4.3 (Multi-index)

An (n-)**multi-index** is an element $\alpha = (\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$. We write $|\alpha| = \alpha_1 + \cdots + \alpha_n$ and $\alpha! = \alpha_1! \cdots \alpha_n!$, and for $x \in \mathbb{R}^n$ we write $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. For $f \in C^k(\mathbb{R}^n)$ with $k \geq |\alpha|$ we write

$$D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Definition 4.4

We say $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ if $f \mathbb{1}_K \in L^p(\mathbb{R}^n)$ for every compact $K \subseteq \mathbb{R}^n$.

Theorem 4.5

Suppose $f \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$ and $g \in C_c^k(\mathbb{R}^n)$ for some $k \geq 0$. Then $f * g \in C^k(\mathbb{R}^n)$, and $D^\alpha(f * g) = f * D^\alpha g$ for every multi-index α with $|\alpha| \leq k$.

Proof. Write $\tau_\xi f(x) = f(x - \xi)$ for translation. Note that $\tau_\xi(f * g) = f * (\tau_\xi g)$ straight from the definition.

Take $k = 0$ first. Choose R so large that $\operatorname{supp} g \subseteq B_{R-1}(0)$. As $|\xi| \to 0$ we have $\tau_\xi g \to g$ pointwise, and $|\tau_\xi g| \leq (\sup |g|) \mathbb{1}_{B_R(0)}$ for $|\xi| < 1$; since f is locally integrable, $|f|(\sup |g|)\mathbb{1}_{B_R(0)}$ is an integrable dominating function on the region that matters. Dominated convergence therefore gives $\tau_\xi(f * g) \to f * g$ pointwise, i.e. $f * g$ is continuous.

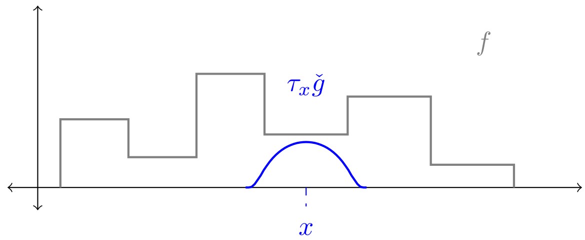
Take $k = 1$. Let $\Delta_i^h g(x) = h^{-1}(g(x + he_i) - g(x))$, so $\Delta_i^h g \to D_i g$ pointwise as $h \to 0$. By the mean value theorem $\Delta_i^h g(x) = D_i g(x + te_i)$ for some $|t| \leq |h|$, so $|\Delta_i^h g| \leq (\sup |D_i g|) \mathbb{1}_{B_R(0)}$ for $|h| < 1$, again a legitimate dominating function.

Hence, by dominated convergence,

$$\frac{(f*g)(x+he_i)-(f*g)(x)}{h} = (f*\Delta_i^h g)(x) \longrightarrow (f*D_i g)(x).$$

So $f*g$ has partial derivatives $f*D_ig$, and these are continuous by the case $k=0$; hence $f*g \in C^1$. Induction on k finishes the proof. $\square$

The picture to have in mind is that of a fixed bump g being slid across f: the value $(f*g)(x)$ is a weighted average of f over a window centred at x, and moving the window moves the average smoothly, no matter how badly behaved f is inside it.



§4.2 Approximate identities

Definition 4.6 (Mollifier)

Fix $\phi \in C_c^\infty(\mathbb{R}^n)$ with $\phi \geq 0$ and $\int_{\mathbb{R}^n} \phi = 1$, supported in the unit ball. For $\varepsilon > 0$ set

$$\phi_\varepsilon(y) = \varepsilon^{-n}\phi(y/\varepsilon).$$

The family $(\phi_\varepsilon)_{\varepsilon>0}$ is called a **mollifier**, or an **approximate identity**.

Each ϕ_ε is smooth, non-negative, supported in $B_\varepsilon(0)$, and has integral 1 (the factor ε^{-n} is exactly what is needed for this). So as $\varepsilon \downarrow 0$ the graphs become taller and narrower, converging in a sense we will make precise later to the Dirac delta.

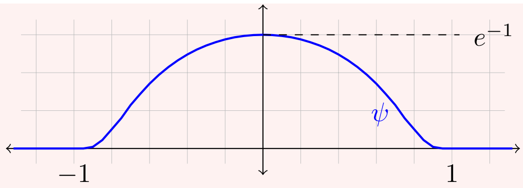
Of course we should check such a ϕ exists, and the standard construction is worth seeing because the same function will reappear when we build test functions in Section **??**.

Example 4.7 (The standard bump)

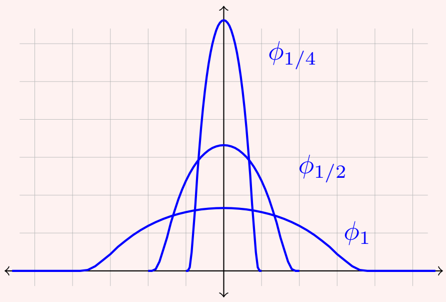
Define $\psi : \mathbb{R} \to \mathbb{R}$ by

$$\psi(x) = \begin{cases} \exp\left(\dfrac{-1}{1-x^2}\right) & |x| < 1, \\ 0 & |x| \geq 1. \end{cases}$$

This is smooth. Away from $x = \pm 1$ this is clear. At $x = \pm 1$, every derivative of ψ on $(-1,1)$ is of the form $R(x)(1-x^2)^{-m}e^{-1/(1-x^2)}$ for a polynomial R and integer m, and $t^{-m}e^{-1/t} \to 0$ as $t \downarrow 0$ since the exponential beats every power. So every derivative extends continuously by 0, and $\psi \in C_c^\infty(\mathbb{R})$ with $\operatorname{supp}\psi = [-1,1]$.



Normalising, $\phi = \psi / \int_{\mathbb{R}} \psi$ is a legitimate mollifier on $\mathbb{R}$; in $\mathbb{R}^n$ one takes $\psi(|x|^2)$ suitably scaled, or simply a product of one-dimensional bumps. The rescaled family ϕ_ε concentrates as $\varepsilon \downarrow 0$:



Each of these has the same area, namely 1.

Before proving that mollification converges, we need one lemma, which says that translation acts continuously on L^p. It is intuitively obvious and genuinely needs proof.

Lemma 4.8 (Continuity of translation)
Let $1 \le p < \infty$ and $g \in L^p(\mathbb{R}^n)$. Then $\|\tau_z g - g\|_{L^p} \to 0$ as $z \to 0$.

Proof. This is a density argument of exactly the shape we shall use again and again. First suppose $g = \mathbb{1}_R$ for a box R. Then $\|\tau_z \mathbb{1}_R - \mathbb{1}_R\|_{L^p}^p = |R \triangle (R+z)| \to 0$ as $z \to 0$, by direct computation with the side lengths. By linearity the result follows for finite disjoint unions of boxes.

Next let B be measurable with $|B| < \infty$. Given $\varepsilon > 0$, Proposition 2.10 gives a finite union of boxes A with $\|\mathbb{1}_A - \mathbb{1}_B\|_{L^p} = |A \triangle B|^{1/p} < \varepsilon$. Since translation is an isometry of L^p,

$$\|\tau_z \mathbb{1}_B - \mathbb{1}_B\|_{L^p} \le \|\tau_z \mathbb{1}_B - \tau_z \mathbb{1}_A\|_{L^p} + \|\tau_z \mathbb{1}_A - \mathbb{1}_A\|_{L^p} + \|\mathbb{1}_A - \mathbb{1}_B\|_{L^p} < 3\varepsilon$$

for $|z|$ small enough. Hence the result holds for indicators of finite-measure sets, so for simple functions of finite support, so – again by the isometry of translation and a three-term estimate – for all of L^p by Proposition 3.12. □

Remark. This fails for $p = \infty$: take $g = \mathbb{1}_{[0,\infty)}$, for which $\|\tau_z g - g\|_{L^\infty} = 1$ for every $z \neq 0$. The failure is the same one that will prevent us from approximating L^∞ functions by continuous ones.

Theorem 4.9 (Mollification)
Let (ϕ_ε) be a mollifier and let $g \in L^p(\mathbb{R}^n)$ with $1 \le p < \infty$. Then $\phi_\varepsilon * g \in C^\infty(\mathbb{R}^n)$,

$\|\phi_\varepsilon * g\|_{L^p} \le \|g\|_{L^p}$, and

$$\phi_\varepsilon * g \to g \quad \text{in } L^p(\mathbb{R}^n) \text{ as } \varepsilon \to 0.$$

Proof. Smoothness is Theorem 4.5: the convolution is defined at every point because $\int_{B_\varepsilon(x)} |g| \le \|g\|_{L^p} \left\|\mathbb{1}_{B_\varepsilon(x)}\right\|_{L^q} < \infty$ by Hölder, so $g \in L^1_{\text{loc}}$, and $\phi_\varepsilon \in C_c^\infty$.

For the convergence, we estimate pointwise. Substituting $y = \varepsilon z$ and using $\int \phi = 1$,

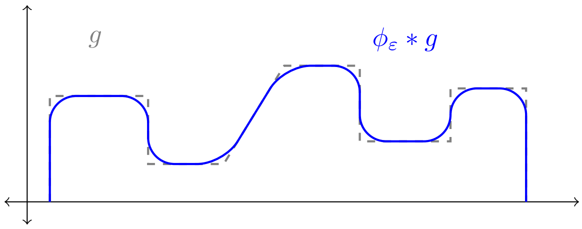
$$\begin{aligned} |(\phi_\varepsilon * g)(x) - g(x)| &= \left| \int_{\mathbb{R}^n} \phi_\varepsilon(y) g(x-y)\, \mathrm{d}y - g(x) \right| \\ &= \left| \int_{\mathbb{R}^n} \phi(z)\big(g(x - \varepsilon z) - g(x)\big)\, \mathrm{d}z \right| \\ &\le \int_{\mathbb{R}^n} \phi(z) \left|\tau_{\varepsilon z} g(x) - g(x)\right|\, \mathrm{d}z. \end{aligned}$$

Now take L^p norms in x and apply Minkowski's integral inequality (Lemma 3.7) to move the norm inside the z-integral:

$$\|\phi_\varepsilon * g - g\|_{L^p} \le \int_{\mathbb{R}^n} \phi(z) \left\|\tau_{\varepsilon z} g - g\right\|_{L^p}\, \mathrm{d}z.$$

The integrand is bounded by $2\phi(z)\|g\|_{L^p} \in L^1$ and tends to 0 pointwise in z as $\varepsilon \to 0$, by Lemma 4.8. So the integral tends to 0 by dominated convergence. The norm bound $\|\phi_\varepsilon * g\|_{L^p} \le \|g\|_{L^p}$ follows from the same application of Minkowski's integral inequality without the subtraction. □

Here is the picture. Convolving a rough function with ϕ_ε rounds off its corners and jumps on the scale ε, and leaves it alone where it was already smooth on that scale.



Corollary 4.10

For $1 \le p < \infty$, the space $C_c^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$.

Proof. Given $g \in L^p$ and $\varepsilon > 0$, first truncate: $g\mathbb{1}_{B_R(0)} \to g$ in L^p as $R \to \infty$ by dominated convergence, so we may assume g has compact support. Now $\phi_\delta * g \to g$ in L^p by Theorem 4.9, and $\phi_\delta * g$ is smooth with support contained in the δ-neighbourhood of $\operatorname{supp} g$, hence compact. □

This is not true for $p = \infty$, and the reason is worth stating: a uniform limit of continuous functions is continuous, so no sequence in $C(\mathbb{R}^n)$ can converge in L^∞ to a function with a genuine jump. The space L^∞ is simply too big – it is not separable either – and this is the first of several places where it will have to be treated separately.

Corollary 4.11

Suppose $f \in L^1_{\mathrm{loc}}(\Omega)$ for $\Omega \subseteq \mathbb{R}^n$ open and

$$\int_\Omega f\varphi \,\mathrm{d}x = 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

Then $f = 0$ almost everywhere.

Proof. Fix a ball B with $\overline{B} \subseteq \Omega$; it suffices to show $f = 0$ a.e. on B. For $x \in B$ and ε smaller than the distance from B to $\partial\Omega$, the function $y \mapsto \phi_\varepsilon(x-y)$ lies in $C_c^\infty(\Omega)$, so the hypothesis gives $(\phi_\varepsilon * f)(x) = 0$. But $f\mathbb{1}_B \in L^1$, so $\phi_\varepsilon * (f\mathbb{1}_B) \to f\mathbb{1}_B$ in L^1 by Theorem 4.9, and for x well inside B these agree. Hence $f = 0$ a.e. on B. □

This innocuous corollary – often called the fundamental lemma of the calculus of variations – is what will allow us to pass from 'the equation holds when tested against every smooth function' to 'the equation holds'. It is used every time we solve a PDE weakly.

§4.3 The Lebesgue differentiation theorem

Mollification says that averages of f over small balls converge to f in L^p. It is natural to ask whether they converge *pointwise*, and the answer is yes, almost everywhere. This is a genuinely non-trivial theorem, and the standard proof introduces a tool – the maximal function – which is the beginning of a whole subject.

Definition 4.12 (Hardy–Littlewood maximal function)

For $f \in L^1(\mathbb{R}^n)$, define $Mf : \mathbb{R}^n \to [0, \infty]$ by

$$Mf(x) = \sup_{r>0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y)| \,\mathrm{d}y.$$

The maximal function is never in L^1 unless $f = 0$ (it decays like $|x|^{-n}$ at infinity, which is not integrable), so the best we can hope for is the following *weak type* bound.

Lemma 4.13

Let $f \in L^1(\mathbb{R}^n)$. Then Mf is measurable and finite almost everywhere, and there is a constant C_n depending only on n with

$$|\{Mf > \lambda\}| \le \frac{C_n \|f\|_{L^1}}{\lambda} \quad \text{for all } \lambda > 0.$$

Proof. Write $A_\lambda = \{Mf > \lambda\}$. For each $x \in A_\lambda$ there is $r_x > 0$ with

$$\frac{1}{|B_{r_x}(x)|} \int_{B_{r_x}(x)} |f| > \lambda.$$

We first check A_λ is open (which gives measurability of Mf, since A_λ is a superlevel set). If $x_m \to x$ with $x \in A_\lambda$, then $\mathbb{1}_{B_{r_x}(x_m)} \to \mathbb{1}_{B_{r_x}(x)}$ almost everywhere and dominated convergence gives

$$\frac{1}{|B_{r_x}(x_m)|}\int_{B_{r_x}(x_m)}|f| \longrightarrow \frac{1}{|B_{r_x}(x)|}\int_{B_{r_x}(x)}|f| > \lambda,$$

so $x_m \in A_\lambda$ for large m. Thus the complement of A_λ is closed under limits, i.e. A_λ is open.

Now let $K \subseteq A_\lambda$ be compact. The balls $\{B_{r_x}(x) : x \in A_\lambda\}$ cover K, so finitely many $B_1, \ldots, B_N$ do. By the Wiener covering lemma there is a disjoint subcollection $B_{i_1}, \ldots, B_{i_k}$ with $\left|\bigcup_{i\le N} B_i\right| \le 3^n \sum_{j\le k}|B_{i_j}|$. Hence

$$|K| \le 3^n \sum_{j=1}^{k} |B_{i_j}| \le \frac{3^n}{\lambda}\sum_{j=1}^{k}\int_{B_{i_j}}|f| \le \frac{3^n}{\lambda}\int_{\mathbb{R}^n}|f| = \frac{3^n \|f\|_{L^1}}{\lambda},$$

using disjointness in the last inequality. Taking the supremum over compact $K \subseteq A_\lambda$ and using inner regularity gives $|A_\lambda| \le 3^n \|f\|_{L^1}/\lambda$. Finally $\{Mf = \infty\} \subseteq A_\lambda$ for every λ, so it is null. □

Theorem 4.14 (Lebesgue differentiation theorem)
Let $f \in L^1(\mathbb{R}^n)$. Then for almost every $x \in \mathbb{R}^n$,

$$\lim_{r\to 0}\frac{1}{|B_r(x)|}\int_{B_r(x)}|f(y) - f(x)|\,\mathrm{d}y = 0.$$

Such x are called **Lebesgue points** of f.

The idea of the proof is a beautiful one and is used throughout analysis: the statement is trivially true for continuous functions, continuous functions are dense, and the maximal inequality controls the error made in passing to the limit. This last ingredient is essential – density alone is never enough for a pointwise statement.

Proof. For $\lambda > 0$ let

$$A_\lambda = \left\{x \in \mathbb{R}^n : \limsup_{r\to 0}\frac{1}{|B_r(x)|}\int_{B_r(x)}|f(y)-f(x)|\,\mathrm{d}y > 2\lambda\right\};$$

it is enough to show $|A_\lambda| = 0$ for every $\lambda > 0$.

Fix $\varepsilon > 0$ and, by Corollary 4.10, choose $g \in C_c^\infty(\mathbb{R}^n)$ with $\|f - g\|_{L^1} < \varepsilon$. For any x and r, the triangle inequality gives

$$\begin{aligned}\frac{1}{|B_r(x)|}\int_{B_r(x)}|f(y)-f(x)|\,\mathrm{d}y \le{}& \frac{1}{|B_r(x)|}\int_{B_r(x)}|f(y)-g(y)|\,\mathrm{d}y \\ &+ \frac{1}{|B_r(x)|}\int_{B_r(x)}|g(y)-g(x)|\,\mathrm{d}y + |g(x)-f(x)|.\end{aligned}$$

The first term is at most $M_{f-g}(x)$ by definition, and the second tends to 0 as $r \to 0$

by continuity of g. So

$$\limsup_{r\to 0} \frac{1}{|B_r(x)|}\int_{B_r(x)} |f(y)-f(x)|\,\mathrm{d}y \le M_{f-g}(x) + |f(x)-g(x)|.$$

Consequently, if $x \in A_\lambda$ then either $M_{f-g}(x) > \lambda$ or $|f(x)-g(x)| > \lambda$. By Lemma 4.13 the first set has measure at most $C_n\varepsilon/\lambda$, and by Chebyshev's inequality the second has measure at most ε/λ. Hence

$$|A_\lambda| \le \frac{(1+C_n)\varepsilon}{\lambda},$$

and since $\varepsilon > 0$ was arbitrary, $|A_\lambda| = 0$. □

Corollary 4.15

For $f \in L^1(\mathbb{R}^n)$ and any mollifier (ϕ_ε), we have $\phi_\varepsilon * f \to f$ almost everywhere as $\varepsilon \to 0$.

Proof. At a Lebesgue point x, bounding $\phi_\varepsilon \le \|\phi\|_{L^\infty}\,\varepsilon^{-n}\mathbb{1}_{B_\varepsilon(0)}$ gives

$$|(\phi_\varepsilon * f)(x) - f(x)| \le \int \phi_\varepsilon(y)|f(x-y)-f(x)|\,\mathrm{d}y \le \frac{C}{|B_\varepsilon(x)|}\int_{B_\varepsilon(x)} |f(y)-f(x)|\,\mathrm{d}y,$$

which tends to 0. □

Corollary 4.16

Let $g \in L^1(\mathbb{R})$ and set $G(x) = \int_{-\infty}^x g(t)\,\mathrm{d}t$. Then G is differentiable at almost every x, with $G'(x) = g(x)$.

Proof. At a Lebesgue point x of g we have, for $h > 0$,

$$\left|\frac{G(x+h)-G(x)}{h} - g(x)\right| = \left|\frac{1}{h}\int_x^{x+h} (g(t)-g(x))\,\mathrm{d}t\right| \le \frac{2}{|B_h(x)|}\int_{B_h(x)} |g-g(x)| \to 0,$$

and similarly for $h < 0$. □

Example 4.17

Take $g = \mathbb{1}_{(0,\infty)}$ on $\mathbb{R}$ (which is in L^1_{loc}, and the theorem applies locally). Every $x \ne 0$ is a Lebesgue point, and indeed

$$G(x) = \int_{-\infty}^x g = \begin{cases} 0 & x \le 0 \\ x & x > 0 \end{cases}$$

is differentiable exactly off 0, with $G' = g$ there. The point $x = 0$ is not a Lebesgue point, and this single bad point is the seed from which the whole theory of distributions grows: we would very much like to say that $G' = g$ and $g' = \delta_0$, and presently we shall.

§5 Hilbert Space and Duality

We turn now to the linear structure. The spaces L^p are Banach spaces, and L^2 is a Hilbert space; the point of this section is to see what that buys us. The headline results are a complete description of the dual space of L^p for $p < \infty$, and – as a by-product of the Hilbert space geometry – a proof of the Radon–Nikodým theorem, which is a theorem of measure theory obtained by functional-analytic means. That last is characteristic of the subject: once the spaces are set up correctly, hard-looking measure theory becomes soft linear algebra.

§5.1 Orthogonal systems of functions

Throughout, $\mathcal{H}$ denotes a Hilbert space over $\mathbb{C}$ with inner product $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable. The model to keep in mind is $\mathcal{H} = L^2(E, \mu)$ with $\langle f, g \rangle = \int_E \bar{f} g \, \mathrm{d}\mu$.

Definition 5.1

A subset $S = \{u_j\}_{j \in J} \subseteq \mathcal{H}$ is **orthogonal** if $\langle u_i, u_j \rangle = 0$ whenever $i \neq j$, and **orthonormal** if in addition $\|u_j\| = 1$ for all j. We say S is **complete** if $\overline{\operatorname{span} S} = \mathcal{H}$, and an orthonormal complete set is an **orthonormal basis**. A Hilbert space is **separable** exactly when it admits a countable orthonormal basis.

If $\{e_n\}_{n \in \mathbb{N}}$ is a countable orthonormal basis then every $x \in \mathcal{H}$ has the expansion

$$x = \sum_{n \in \mathbb{N}} \langle e_n, x \rangle e_n, \qquad \|x\|^2 = \sum_{n \in \mathbb{N}} |\langle e_n, x \rangle|^2,$$

the series converging in norm; this is Parseval's identity, and it says that $\mathcal{H} \cong \ell^2$ as soon as $\mathcal{H}$ is separable and infinite-dimensional. Abstractly, then, there is only one separable infinite-dimensional Hilbert space. What makes the subject non-trivial is that the *same* space carries many different natural bases, and that choosing a good one is often the whole problem.

Example 5.2

Some orthonormal bases worth knowing.

(i) The trigonometric system $\{x \mapsto e^{-2\pi i n x} : n \in \mathbb{Z}\}$ is an orthonormal basis of $L^2((0,1))$. Orthonormality is a computation; completeness follows either from Stone–Weierstrass (the trigonometric polynomials are dense in the continuous periodic functions in the uniform norm) or from Corollary 4.10. This is the classical theory of Fourier series, in one line.

(ii) Let

$$\psi(x) = \begin{cases} 1 & 0 \leq x < 1/2, \\ -1 & 1/2 \leq x < 1, \\ 0 & \text{otherwise}, \end{cases}$$

and set $\psi_{n,k}(x) = 2^{n/2} \psi(2^n x - k)$. Then $\{\psi_{n,k} : n, k \in \mathbb{Z}\}$ is an orthonormal basis of $L^2(\mathbb{R})$, the **Haar system**. It is the simplest example of a wavelet basis, and unlike the trigonometric system its elements are localised in space as well as in frequency.

(iii) Let $\mathcal{H} = L^2(\mathbb{R}, e^{-x^2}\, dx)$, with inner product $\langle f, g\rangle = \int_{\mathbb{R}} \overline{f} g e^{-x^2}\, dx$. Applying Gram–Schmidt to the linearly independent family $1, x, x^2, \ldots$ produces orthogonal polynomials H_k with $\deg H_k = k$; normalised so that the coefficient of x^k in H_k is 2^k, these are the **Hermite polynomials**, and they form a complete orthogonal set.

The result from Linear Analysis we shall use over and over is the following.

Theorem 5.3 (Riesz representation, Hilbert space form)
Let $\mathcal{H}$ be a Hilbert space and $\Lambda : \mathcal{H} \to \mathbb{C}$ a bounded linear functional. Then there is a unique $w \in \mathcal{H}$ with

$$\Lambda u = \langle w, u\rangle \quad \text{for all } u \in \mathcal{H},$$

and $\|\Lambda\|_{\mathcal{H}'} = \|w\|$.

Proof. We recall the argument. If $\Lambda = 0$ take $w = 0$. Otherwise $K = \ker \Lambda$ is a proper closed subspace, so there is $v \in K^\perp$ with $\|v\| = 1$ (take any $z \notin K$ and subtract its orthogonal projection onto K, which exists because K is closed and convex). For any u, the vector $u - \frac{\Lambda u}{\Lambda v} v$ lies in K, hence is orthogonal to v, so $\langle v, u\rangle = \frac{\Lambda u}{\Lambda v}$. Thus $w = \overline{\Lambda v} v$ works. Uniqueness and the norm identity are immediate. □

§5.2 The Radon–Nikodým theorem

Here is a first illustration of the power of Theorem 5.3.

Definition 5.4
Let μ, ν be measures on a measurable space $(E, \mathcal{E})$. We say ν is **absolutely continuous** with respect to μ, written $\nu \ll \mu$, if $\mu(A) = 0 \implies \nu(A) = 0$ for all $A \in \mathcal{E}$. We say μ and ν are **mutually singular**, written $\mu \perp \nu$, if there is a measurable Z with $\mu(Z) = 0 = \nu(Z^c)$.

Theorem 5.5 (Radon–Nikodým)
Let μ, ν be finite measures on $(E, \mathcal{E})$ with $\nu \ll \mu$. Then there is a non-negative $w \in L^1(E, \mu)$ such that

$$\nu(A) = \int_A w\, d\mu \quad \text{for all } A \in \mathcal{E};$$

equivalently, $\int_E F\, d\nu = \int_E F w\, d\mu$ for every measurable $F : E \to [0, \infty]$.

Proof. The trick is to introduce two auxiliary measures which are comparable to both μ and ν. Set $\alpha = \mu + 2\nu$ and $\beta = \nu + 2\mu$, both finite. Since $\mu \le \alpha$ and $\nu \le \alpha$, we have

$$\mathcal{H} := L^2(E, \alpha) \subseteq L^2(E, \mu) \cap L^2(E, \nu),$$

and every $f \in \mathcal{H}$ is β-integrable. Consider the linear functional

$$\Lambda(f) = \int_E f \, \mathrm{d}\beta, \qquad f \in \mathcal{H}.$$

It is bounded, since

$$|\Lambda(f)| \le \int_E |f| \, \mathrm{d}\beta = 2\int_E |f| \, \mathrm{d}\mu + \int_E |f| \, \mathrm{d}\nu \le 2\int_E |f| \, \mathrm{d}\alpha \le 2\sqrt{\alpha(E)} \, \|f\|_{L^2(\alpha)}$$

by Cauchy–Schwarz. By Theorem 5.3 there is $g \in \mathcal{H}$ with

$$\int_E f \, \mathrm{d}\beta = \int_E gf \, \mathrm{d}\alpha \quad \text{for all } f \in \mathcal{H}.$$

Unwinding the definitions of α and β, this says

$$\int_E (2g-1)f \, \mathrm{d}\nu = \int_E (2-g)f \, \mathrm{d}\mu \quad \text{for all } f \in \mathcal{H}. \tag{$*$}$$

We claim $\frac{1}{2} \le g \le 2$ almost everywhere with respect to both μ and ν. Indeed, let $A_j = \{g < \frac{1}{2} - \frac{1}{j}\}$ and put $f = \mathbb{1}_{A_j}$ in $(*)$:

$$\frac{3}{2}\mu(A_j) \le \int_E (2-g)\mathbb{1}_{A_j} \, \mathrm{d}\mu = \int_E (2g-1)\mathbb{1}_{A_j} \, \mathrm{d}\nu \le -\frac{2}{j}\nu(A_j) \le 0,$$

forcing $\mu(A_j) = 0$ and then $\nu(A_j) = 0$. Letting $j \to \infty$ gives $g \ge 1/2$ a.e.; the argument at the other end, with $A_j = \{g > 2 + 1/j\}$, gives $g \le 2$ a.e.

Now let $Z = \{g = 1/2\}$. Putting $f = \mathbb{1}_Z$ in $(*)$ gives $0 = \int_Z (2-g) \, \mathrm{d}\mu \ge \frac{3}{2}\mu(Z)$, so $\mu(Z) = 0$, and hence $\nu(Z) = 0$ by absolute continuity. Given measurable $F : E \to [0,\infty]$, define on Z^c

$$f = \frac{F}{2g-1}, \qquad w = \frac{2-g}{2g-1},$$

and $f = w = 0$ on Z. Both are well defined and bounded multiples of F and 1 respectively, since $2g-1 \in [0,3]$ is bounded away from 0 off Z after a further exhaustion, and applying $(*)$ (first for bounded F, then in general by monotone convergence) gives

$$\int_E F \, \mathrm{d}\nu = \int_{Z^c} (2g-1)f \, \mathrm{d}\nu = \int_{Z^c} (2-g)f \, \mathrm{d}\mu = \int_E Fw \, \mathrm{d}\mu$$

as required. Taking $F = 1$ shows $w \in L^1(E,\mu)$, and taking $F = \mathbb{1}_A$ gives the stated form. □

§5.3 Dual spaces

Definition 5.6 (Dual space)

Let X be a topological vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The **(topological) dual** X' is the vector space of continuous linear functionals $\Lambda : X \to \mathbb{F}$. If X is normed, we norm X' by

$$\|\Lambda\|_{X'} = \sup_{\|x\|_X \le 1} |\Lambda(x)|.$$

The space X' is always a Banach space, complete regardless of whether X is, since completeness is inherited from the target field. What is not at all obvious is that X' contains anything: the definition alone does not produce a single non-zero functional. That is the business of the Hahn–Banach theorem, which we prove in Section 6; for now we record what we shall need.

Lemma 5.7

Let X be a normed space and $x \neq y$ in X. Then there is $\Lambda \in X'$ with $\Lambda(x) \neq \Lambda(y)$.

We defer the proof to Corollary 6.9. Granting it, for each $x \in X$ the evaluation map $f_x : X' \to \mathbb{F}$, $\Lambda \mapsto \Lambda(x)$, is linear and bounded (with $|f_x(\Lambda)| \leq \|x\|\,\|\Lambda\|$), hence $f_x \in X'' = (X')'$; and Lemma 5.7 says $x \mapsto f_x$ is injective. In fact it is isometric, again by Hahn–Banach.

Definition 5.8 (Reflexive)

A Banach space X is **reflexive** if the canonical map $X \to X''$, $x \mapsto f_x$, is surjective (hence an isometric isomorphism), in which case we write $X = X''$.

For a Hilbert space, Theorem 5.3 gives a conjugate-linear isometry $\mathcal{H} \to \mathcal{H}'$; applying it twice shows that every Hilbert space is reflexive. This will matter enormously later: reflexivity is what makes bounded sequences have weakly convergent subsequences.

§5.4 The dual of L^p

Fix a measure space $(E, \mathcal{E}, \mu)$ and conjugate exponents p, q. Hölder's inequality says that for $g \in L^q$ the map

$$\Lambda_g(f) = \int_E gf \,\mathrm{d}\mu$$

is a bounded linear functional on L^p with $\|\Lambda_g\|_{(L^p)'} \leq \|g\|_{L^q}$. In fact equality holds.

Lemma 5.9

For $1 \leq p \leq \infty$ and $g \in L^q$ we have $\|\Lambda_g\|_{(L^p)'} = \|g\|_{L^q}$. Hence $\kappa : L^q \to (L^p)'$, $g \mapsto \Lambda_g$, is a linear isometry.

Proof. Only '$\geq$' needs proof. Suppose $1 < q < \infty$ and $g \neq 0$; take

$$f = \frac{|g|^{q-1}\overline{\operatorname{sgn} g}}{\|g\|_{L^q}^{q-1}}, \quad \text{so that} \quad \|f\|_{L^p}^p = \frac{\int |g|^{(q-1)p}}{\|g\|_{L^q}^{(q-1)p}} = 1$$

since $(q-1)p = q$. Then $\Lambda_g(f) = \|g\|_{L^q}$. For $q = 1$ take $f = \overline{\operatorname{sgn} g}$, of L^∞ norm 1. For $q = \infty$, given $\varepsilon > 0$ let $A = \{|g| > \|g\|_{L^\infty} - \varepsilon\}$, which has positive measure; assuming μ σ-finite we may shrink it to have finite positive measure, and $f = \mu(A)^{-1}\mathbb{1}_A\overline{\operatorname{sgn} g}$ has $\|f\|_{L^1} = 1$ and $\Lambda_g(f) \geq \|g\|_{L^\infty} - \varepsilon$. $\square$

So L^q sits isometrically inside $(L^p)'$. The theorem is that for $p < \infty$ it is everything.

Theorem 5.10

Let $(E, \mathcal{E}, \mu)$ be σ-finite, let $1 \leq p < \infty$ and let q be conjugate to p. Then $\kappa : L^q(E,\mu) \to L^p(E,\mu)'$ is an isometric isomorphism. We write $L^p(E,\mu)' = L^q(E,\mu)$.

Proof. By Lemma 5.9 it remains to prove surjectivity, and by Lemma 5.9 again it is enough to produce, for each $\Lambda \in (L^p)'$, some $g \in L^q$ with $\Lambda = \Lambda_g$.

Reduction to real scalars and positive functionals. Writing $f = f_r + i f_i$ and $\Lambda_r(f) = \operatorname{Re}\Lambda(f)$, $\Lambda_i(f) = \operatorname{Im}\Lambda(f)$ for real f, a complex functional is determined by a pair of real ones, so we may take the scalars real. A real-linear bounded $\Lambda : L^p(E;\mathbb{R}) \to \mathbb{R}$ is called **positive** if $\Lambda(f) \geq 0$ whenever $f \geq 0$; one checks (this is on the example sheet, and is the Jordan decomposition in disguise) that every bounded real-linear functional is a difference $\Lambda = \Lambda^+ - \Lambda^-$ of positive bounded ones. So assume Λ positive.

The finite measure case. Suppose first $\mu(E) < \infty$. Define $\nu(A) = \Lambda(\mathbb{1}_A)$ for $A \in \mathcal{E}$. Then $\nu \geq 0$ by positivity, $\nu(\emptyset) = 0$, and ν is finitely additive by linearity. It is countably additive: if $B = \bigcup_{n\geq 1} A_n$ with the A_n disjoint, and $B_k = \bigcup_{n \leq k} A_n$, then

$$\|\mathbb{1}_B - \mathbb{1}_{B_k}\|_{L^p} = \mu(B \setminus B_k)^{1/p} \to 0,$$

so $\nu(B) - \nu(B_k) = \Lambda(\mathbb{1}_B - \mathbb{1}_{B_k}) \to 0$ by continuity of Λ. The same computation shows $\mu(A) = 0 \implies \nu(A) = 0$, i.e. $\nu \ll \mu$. By Theorem 5.5 there is a non-negative $g \in L^1(E,\mu)$ with $\Lambda(\mathbb{1}_A) = \int_A g \,\mathrm{d}\mu$ for all A; by linearity $\Lambda(f) = \int_E fg \,\mathrm{d}\mu$ for all simple f, and hence for all bounded measurable f by dominated convergence.

g lies in L^q. Suppose $1 < p < \infty$. For $N > 0$ let $g_N = \min(g, N)$ and $f_N = g_N^{q-1}$, a bounded function. Then

$$\int_E g_N^q \,\mathrm{d}\mu \leq \int_E f_N g \,\mathrm{d}\mu = \Lambda(f_N) \leq \|\Lambda\|\,\|f_N\|_{L^p} = \|\Lambda\| \left(\int_E g_N^q \,\mathrm{d}\mu\right)^{1/p},$$

using $(q-1)p = q$. Dividing (the integral is finite as g_N is bounded and $\mu(E) < \infty$) gives $\|g_N\|_{L^q} \leq \|\Lambda\|$, and monotone convergence as $N \to \infty$ gives $\|g\|_{L^q} \leq \|\Lambda\| < \infty$. For $p = 1$ one argues instead that $\mu(\{g > \|\Lambda\| + \varepsilon\}) = 0$, directly from $\Lambda(\mathbb{1}_A) \leq \|\Lambda\|\,\mu(A)$, so $g \in L^\infty$. Finally, since $g \in L^q$ the functional Λ_g is bounded on L^p and agrees with Λ on the dense set of simple functions (Proposition 3.12), so $\Lambda = \Lambda_g$.

The σ-finite case. Write $E = \bigcup_k E_k$ with $E_k \uparrow$ and $\mu(E_k) < \infty$. Applying the above on each E_k produces $g_k \in L^q(E_k)$ representing Λ on functions supported in E_k; by uniqueness these are consistent, so they patch to a measurable g on E with $\|g\mathbb{1}_{E_k}\|_{L^q} \leq \|\Lambda\|$ for every k. Monotone convergence gives $g \in L^q$ with $\|g\|_{L^q} \leq \|\Lambda\|$, and $\Lambda = \Lambda_g$ on the dense set of functions supported in some E_k, hence everywhere. □

Corollary 5.11

For $1 < p < \infty$, $L^p(\mathbb{R}^n)$ is reflexive.

Proof. $(L^p)'' = (L^q)' = L^p$, and one checks that the composite isomorphism is the canonical embedding. □

The theorem genuinely fails at $p = \infty$: the dual of L^∞ is much larger than L^1. To see this, note that $C([0,1]) \subseteq L^\infty([0,1])$ and $f \mapsto f(0)$ is a bounded functional on $C([0,1])$; Hahn–Banach extends it to L^∞, and no $g \in L^1$ can represent it (it vanishes on every f vanishing near 0, which forces $g = 0$ a.e., but the functional is not zero). Correspondingly neither L^1 nor L^∞ is reflexive. This is not a technicality to be regretted: it is why the natural home for many variational problems is L^p with $1 < p < \infty$, and why problems formulated in L^1 (or in the space of measures) are genuinely harder.

§5.5 The Riesz representation theorem for measures

There is one more duality worth recording, and it goes the other way: instead of describing functionals in terms of functions, it describes them in terms of *measures*. This is the sense in which a measure 'is' a way of integrating continuous functions.

Definition 5.12

Let E be a topological space and $\mathcal{E} \supseteq \mathcal{B}(E)$ a σ-algebra. A measure μ on $(E, \mathcal{E})$ is **regular** if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are a closed C and an open O with $C \subseteq A \subseteq O$ and $\mu(O \setminus C) < \varepsilon$.

Lebesgue measure on $(\mathbb{R}^n, \mathcal{M})$ is regular, by Theorem 2.5(iii). Now if μ is a finite regular measure on $\mathbb{R}^n$, then every $f \in C_c^0(\mathbb{R}^n; \mathbb{R})$ is measurable and bounded, so

$$\Lambda(f) = \int_{\mathbb{R}^n} f \, \mathrm{d}\mu$$

defines a positive bounded linear functional on $C_c^0(\mathbb{R}^n; \mathbb{R})$. The question is whether μ can be recovered from Λ, and whether every such Λ arises this way.

Lemma 5.13

A finite regular measure μ on $\mathbb{R}^n$ is determined by the functional $\Lambda(f) = \int f \, \mathrm{d}\mu$ on $C_c^0(\mathbb{R}^n; \mathbb{R})$.

Proof. We would like to take $f = \mathbb{1}_A$, but indicators are not continuous, so we approximate. Let O be open, and for $k \in \mathbb{N}$ set $O_k = O \cap \{|x| < k\}$ and

$$\chi_k(x) = \begin{cases} 1 & x \in O_k,\ \operatorname{dist}(x, O_k^c) \geq 1/k, \\ k \operatorname{dist}(x, O_k^c) & x \in O_k,\ \operatorname{dist}(x, O_k^c) < 1/k, \\ 0 & x \notin O_k. \end{cases}$$

Each χ_k is continuous with compact support (distance functions are 1-Lipschitz) and $\chi_k \uparrow \mathbb{1}_O$ pointwise. By monotone convergence $\Lambda(\chi_k) \to \mu(O)$. So Λ determines μ on open sets, and hence on all of $\mathcal{E}$ by regularity. ☐

Theorem 5.14 (Riesz–Markov)

Let $\Lambda : C_c^0(\mathbb{R}^n; \mathbb{R}) \to \mathbb{R}$ be a positive bounded linear functional. Then there is a

σ-algebra $\mathcal{M} \supseteq \mathcal{B}(\mathbb{R}^n)$ and a unique finite regular measure μ on it with

$$\Lambda(f) = \int_{\mathbb{R}^n} f \, \mathrm{d}\mu \quad \text{for all } f \in C_c^0(\mathbb{R}^n; \mathbb{R}).$$

The proof is a construction of μ from Λ by an outer measure argument – one defines $\mu(O) = \sup\{\Lambda(f) : 0 \le f \le 1, \operatorname{supp} f \subseteq O\}$ for open O and extends by regularity – and it is genuinely technical. We shall take it on trust.

Example 5.15

Take $\Lambda(f) = f(0)$. This is positive and bounded, and the measure it represents is $\mu = \delta_0$, the Dirac point mass at the origin. The reader who feels that δ_0 'should' be a function will be pleased to know that in Section **??** it becomes a perfectly respectable distribution, and that this example is the prototype.

§6 Weak Topologies and Compactness

In finite dimensions, the Bolzano–Weierstrass theorem says that bounded sequences have convergent subsequences, and this single fact underlies most existence proofs in analysis: to produce an object with a desired property, construct a bounded sequence of approximations and pass to a limit. In infinite dimensions this fails completely. In ℓ^2 the orthonormal sequence (e_n) is bounded but $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$, so no subsequence is Cauchy; indeed the closed unit ball of a normed space is compact if and only if the space is finite-dimensional.

The remedy is to keep the sequence and change the topology. If we declare fewer sets to be open, more sets become compact, and with luck we can weaken the topology enough to restore Bolzano–Weierstrass while keeping enough separation to make the limits useful. This section carries that out. We need the Hahn–Banach theorem first, because a weak topology is built from linear functionals and is useless unless there are plenty of them.

§6.1 The Hahn–Banach theorem

Hahn–Banach is an extension theorem: a bounded functional defined on a subspace extends to the whole space with the same bound. We work over $\mathbb{R}$; the complex case follows by taking real parts, since a complex-linear Λ is determined by $\ell = \operatorname{Re}\Lambda$ via $\Lambda(x) = \ell(x) - i\ell(ix)$.

For the sake of generality we replace the norm bound by something weaker.

Definition 6.1 (Sublinear functional)

A **sublinear functional** on a real vector space X is a map $p : X \to \mathbb{R}$ with

$$p(x+y) \le p(x) + p(y), \qquad p(tx) = tp(x)$$

for all $x, y \in X$ and $t \ge 0$.

Example 6.2

Every seminorm is sublinear, and so is $|\ell|$ for any linear ℓ. But sublinear functionals need not be non-negative or symmetric: on $\mathbb{R}$, $p(x) = x$ is sublinear.

Note that a one-sided bound $\ell(x) \leq p(x)$ automatically gives a two-sided one, since $-\ell(x) = \ell(-x) \leq p(-x)$, whence $-p(-x) \leq \ell(x) \leq p(x)$.

Lemma 6.3 (Banach extension lemma)

Let X be a real vector space, p a sublinear functional on X, $M \leq X$ a subspace, and $\ell : M \to \mathbb{R}$ linear with $\ell \leq p$ on M. Then for any $x \in X \setminus M$ there is a linear $\widetilde{\ell}$ on $\widetilde{M} = \operatorname{span}(M \cup \{x\})$ extending ℓ with $\widetilde{\ell} \leq p$ on $\widetilde{M}$.

Proof. Every $w \in \widetilde{M}$ is uniquely $w = \lambda x + y$ with $y \in M$, $\lambda \in \mathbb{R}$, so $\widetilde{\ell}$ is determined by the single number $a = \widetilde{\ell}(x)$, and the problem is to choose a compatibly.

For $y, z \in M$ we have

$$\ell(y) + \ell(z) = \ell(y + z) \leq p(y + z) \leq p(y - x) + p(z + x),$$

so that $\ell(y) - p(y - x) \leq p(z + x) - \ell(z)$ for all $y, z \in M$. Hence

$$a := \sup_{y \in M} \big(\ell(y) - p(y - x)\big) \leq \inf_{z \in M} \big(p(z + x) - \ell(z)\big) < \infty,$$

and both bounds are finite. By construction $\ell(y) - a \leq p(y - x)$ and $\ell(z) + a \leq p(z + x)$ for all $y, z \in M$. Now take $w = \lambda x + y$. For $\lambda > 0$, apply the second inequality with $z = \lambda^{-1} y$ and multiply by λ; for $\lambda < 0$, apply the first with $y' = -\lambda^{-1} y$ and multiply by $-\lambda$; for $\lambda = 0$ there is nothing to prove. In each case we get $\widetilde{\ell}(w) \leq p(w)$. □

If X/M is finite-dimensional we can now finish by induction, and if X is separable we can finish by a countable induction plus a limiting argument. In general we need a form of the axiom of choice.

Proposition 6.4 (Zorn's lemma)

Let $(S, \leq)$ be a non-empty partially ordered set in which every totally ordered subset has an upper bound. Then S has a maximal element.

Theorem 6.5 (Hahn–Banach)

Let X be a real vector space, p a sublinear functional on X, $M \leq X$ and $\ell : M \to \mathbb{R}$ linear with $\ell \leq p$ on M. Then there is a linear $\widetilde{\ell} : X \to \mathbb{R}$ extending ℓ with $\widetilde{\ell} \leq p$ on X.

Proof. Let S be the set of pairs (N, ℓ^*) where $M \leq N \leq X$ and $\ell^* : N \to \mathbb{R}$ is linear, extends ℓ, and satisfies $\ell^* \leq p$ on N. Order S by $(N_1, \ell_1^*) \leq (N_2, \ell_2^*)$ if $N_1 \leq N_2$ and ℓ_2^* extends ℓ_1^*. Then $S \neq \emptyset$ (it contains (M, ℓ)), and any chain $T \subseteq S$ has the upper bound (N_T, ℓ_T) with $N_T = \bigcup_{(N,\ell^*) \in T} N$ – a subspace because T is totally ordered – and ℓ_T the common value. By Proposition 6.4 there is a maximal $(\widetilde{N}, \widetilde{\ell})$;

and if $\widetilde{N} \neq X$ then Lemma 6.3 would extend it further, contradicting maximality. So $\widetilde{N} = X$. $\square$

Corollary 6.6

Let X be a normed space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $M \leq X$, and $\Lambda : M \to \mathbb{F}$ a bounded linear functional. Then there is $\widetilde{\Lambda} \in X'$ extending Λ with $\left\|\widetilde{\Lambda}\right\|_{X'} = \|\Lambda\|_{M'}$.

Proof. For $\mathbb{F} = \mathbb{R}$ apply Theorem 6.5 with $p(x) = \|\Lambda\|_{M'} \|x\|$. For $\mathbb{F} = \mathbb{C}$, write $\Lambda(x) = \ell(x) - i\ell(ix)$ with $\ell = \operatorname{Re}\Lambda$ real-linear; since for each x we may pick θ with $|\Lambda(x)| = \ell(e^{i\theta}x)$, the real and complex norms agree, and extending ℓ then setting $\widetilde{\Lambda}(x) = \widetilde{\ell}(x) - i\widetilde{\ell}(ix)$ does the job. $\square$

Corollary 6.7

For any $x \neq 0$ in a normed space X there is $\Lambda \in X'$ with $\|\Lambda\|_{X'} = 1$ and $\Lambda(x) = \|x\|$. Consequently $\|x\| = \sup_{\|\Lambda\|_{X'} \leq 1} |\Lambda(x)|$, and the canonical map $X \to X''$ is isometric.

Proof. Define $\Lambda(tx) = t\,\|x\|$ on $M = \mathbb{F}x$, of norm 1, and extend by Corollary 6.6. $\square$

§6.2 Geometric forms

Hahn–Banach has a geometric face which is often more useful. Recall that $A \subseteq X$ is **convex** if $tx + (1-t)y \in A$ for all $x, y \in A$ and $t \in [0,1]$. In $\mathbb{R}^n$ one knows that disjoint compact convex sets can be separated by a hyperplane; Hahn–Banach extends this to infinite dimensions, with the linear functional playing the role of the hyperplane.

Theorem 6.8 (Geometric Hahn–Banach)

Let A, B be disjoint, non-empty, convex subsets of a real or complex Banach space X.

(a) If A is open, there are $\Lambda \in X'$ and $\gamma \in \mathbb{R}$ with

$$\operatorname{Re}\Lambda(x) < \gamma \leq \operatorname{Re}\Lambda(y) \quad \text{for all } x \in A, y \in B.$$

If B is also open, the second inequality is strict too.

(b) If A is compact and B is closed, there are $\Lambda \in X'$ and $\gamma_1 < \gamma_2$ with

$$\operatorname{Re}\Lambda(x) < \gamma_1 < \gamma_2 < \operatorname{Re}\Lambda(y) \quad \text{for all } x \in A, y \in B.$$

Proof. We may take X real.

(a) Pick $a_0 \in A$, $b_0 \in B$ and set $x_0 = b_0 - a_0$, and let $C = A - B + x_0$. Then C is convex (a sum of convex sets), open (a union of translates of the open A), and contains 0; and $x_0 \notin C$ since $A \cap B = \emptyset$. Define the **Minkowski gauge**

$$p(x) = \inf\{t > 0 : t^{-1}x \in C\}.$$

One checks (example sheet) that p is sublinear, that $p(x) \le k\,\|x\|$ for some $k > 0$ (because C contains a ball about 0), and that $p(y) < 1$ for $y \in C$; since $x_0 \notin C$ we get $p(x_0) \ge 1$.

On $M = \mathbb{R}x_0$ define $f(tx_0) = t$. Then $f \le p$ on M: for $t > 0$, $f(tx_0) = t \le tp(x_0) = p(tx_0)$, and for $t \le 0$, $f(tx_0) = t \le 0 \le p(tx_0)$. By Theorem 6.5 extend f to a linear Λ on X with $\Lambda \le p$; then $-k\,\|x\| \le -p(-x) \le \Lambda(x) \le p(x) \le k\,\|x\|$, so $\Lambda \in X'$.

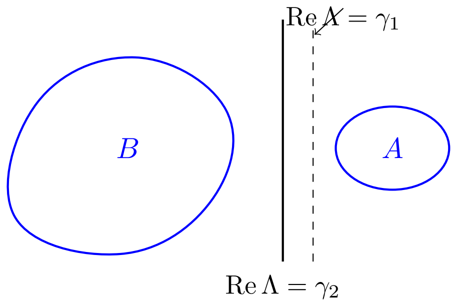
For $a \in A$, $b \in B$ we have $a - b + x_0 \in C$, so

$$\Lambda(a) - \Lambda(b) + 1 = \Lambda(a - b + x_0) \le p(a - b + x_0) < 1,$$

giving $\Lambda(a) < \Lambda(b)$. Thus $\Lambda(A)$ and $\Lambda(B)$ are convex subsets of $\mathbb{R}$ (that is, intervals) with $\sup \Lambda(A) \le \inf \Lambda(B)$. As Λ is a non-zero continuous linear map it is open, so $\Lambda(A)$ is an open interval and $\gamma = \sup \Lambda(A)$ is not attained; this is the claim.

(b) Since A is compact, B closed and they are disjoint, $d = \inf\{\|a - b\| : a \in A, b \in B\} > 0$. Let $V = B_{d/2}(0)$; then $A + V$ is open, convex, and disjoint from B, so (a) provides Λ with $\sup \Lambda(A + V) \le \inf \Lambda(B)$. Since $\Lambda(A)$ is a compact subset of the open set $\Lambda(A + V)$ we may insert the two constants $\gamma_1 < \gamma_2$. $\square$

Pictorially: a point outside a closed convex set can be cut off from it by a hyperplane, with room to spare.



Corollary 6.9

For distinct x, y in a normed space X there is $\Lambda \in X'$ with $\Lambda(x) \neq \Lambda(y)$.

Proof. Apply Theorem 6.8(b) with $A = \{x\}$ and $B = \{y\}$ (or Corollary 6.7 to $x - y$). $\square$

Corollary 6.10

Let $M \le X$ be a subspace with $\overline{M} \neq X$, and let $x_0 \in X \setminus \overline{M}$. Then there is $\Lambda \in X'$ with $\Lambda(x_0) = 1$ and $\Lambda|_M = 0$.

Proof. Take $A = \{x_0\}$, $B = \overline{M}$ in Theorem 6.8(b) to get Λ_0 with $\operatorname{Re}\Lambda_0(x_0) < \gamma_1 < \gamma_2 < \operatorname{Re}\Lambda_0(y)$ for $y \in \overline{M}$. Since $\overline{M}$ is a subspace, $\Lambda_0(\overline{M})$ is a subspace of $\mathbb{F}$ bounded away from $\operatorname{Re}\Lambda_0(x_0)$, which forces $\Lambda_0|_M = 0$; then rescale. $\square$

Remark. Corollary 6.10 is the standard tool for proving *density*: to show a subspace M is dense in X, it suffices to show that the only $\Lambda \in X'$ vanishing on M is $\Lambda = 0$. This is usually much easier, because we have a concrete description of X' to work with. We shall use exactly this manoeuvre when studying Sobolev spaces.

§6.3 Consequences of Baire category

We record, without proof, three theorems from Linear Analysis which all follow from the Baire category theorem (a complete metric space is not a countable union of nowhere dense sets). Their common feature is that they turn a pointwise or algebraic hypothesis into a uniform or topological conclusion.

Theorem 6.11 (Principle of uniform boundedness)

Let X be a Banach space, Y a normed space, and $\mathcal{T} \subseteq \mathcal{B}(X, Y)$ a family of bounded operators with $\sup_{T\in\mathcal{T}} \|Tx\| < \infty$ for every $x \in X$. Then $\sup_{T\in\mathcal{T}} \|T\| < \infty$.

Theorem 6.12 (Open mapping theorem)

A surjective bounded linear map between Banach spaces is open. In particular a bijective bounded linear map has bounded inverse.

Theorem 6.13 (Closed graph theorem)

A linear map between Banach spaces whose graph is closed is bounded.

The consequence we need immediately is the following, which says that weak convergence – defined in a moment – cannot produce unbounded sequences.

Proposition 6.14

Let X be a Banach space and (x_n) a sequence with $\Lambda(x_n)$ convergent for every $\Lambda \in X'$. Then $(\|x_n\|)$ is bounded.

Proof. View each x_n as an element f_{x_n} of $X'' = \mathcal{B}(X', \mathbb{F})$, noting X' is a Banach space. For each Λ the sequence $f_{x_n}(\Lambda) = \Lambda(x_n)$ is convergent, hence bounded. By Theorem 6.11, $\sup_n \|f_{x_n}\|_{X''} < \infty$; and $\|f_{x_n}\|_{X''} = \|x_n\|$ by Corollary 6.7. $\square$

§6.4 Seminorms and locally convex spaces

To define the weak topologies properly we need topologies generated not by a norm but by a family of seminorms.

Definition 6.15 (Seminorm)

A **seminorm** on a vector space X over $\mathbb{F}$ is a map $p : X \to \mathbb{R}$ with $p(x + y) \leq p(x) + p(y)$ and $p(\lambda x) = |\lambda| p(x)$.

Taking $\lambda = 0$ gives $p(0) = 0$, and then $0 = p(0) \leq 2p(x)$ shows $p \geq 0$. So a seminorm is a norm except that $p(x) = 0$ need not force $x = 0$. A single seminorm is therefore too coarse to separate points; the idea is to use many of them.

Definition 6.16

A family $\mathcal{P}$ of seminorms on X is **separating** if for every $x \neq 0$ there is $p \in \mathcal{P}$ with $p(x) \neq 0$.

Given a separating family $\mathcal{P}$, set $V(p,n) = \{x : p(x) < 1/n\}$ for $p \in \mathcal{P}$ and $n \in \mathbb{N}$, let $\dot{\beta}$ be the collection of finite intersections of such sets, and let $\beta = \{x + B : x \in X, B \in \dot{\beta}\}$.

Theorem 6.17

β is a base for a Hausdorff topology $\tau_{\mathcal{P}}$ on X, with respect to which addition, scalar multiplication and every $p \in \mathcal{P}$ are continuous.

Definition 6.18

A space of the form $(X, \tau_{\mathcal{P}})$ is a **locally convex topological vector space**, or LCTVS. If $\mathcal{P} = \{p_n\}_{n \in \mathbb{N}}$ is countable, $\tau_{\mathcal{P}}$ is metrisable by

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \frac{p_n(x-y)}{1 + p_n(x-y)},$$

and if this metric is complete we call X a **Fréchet space**.

'Locally convex' refers to the fact that the basic neighbourhoods $V(p,n)$ are convex, which is exactly the hypothesis needed for Hahn–Banach separation arguments to work. All the spaces in this course are locally convex; we shall meet Fréchet spaces again when we build test functions.

§6.5 Weak and weak-$*$ topologies

Let X be a Banach space. Taking $\mathcal{P} = \{\|\cdot\|\}$ recovers the norm topology, which for contrast we now call the **strong topology** τ_s; convergence in it, written $x_n \to x$, means $\|x_n - x\| \to 0$.

Definition 6.19 (Weak topology)

For $\Lambda \in X'$ the map $p_\Lambda(x) = |\Lambda(x)|$ is a seminorm, and the family $\{p_\Lambda : \Lambda \in X'\}$ is separating by Corollary 6.9. The topology it generates is the **weak topology** τ_w on X. Thus $x_n \rightharpoonup x$ ('x_n converges weakly to x') if and only if

$$\Lambda(x_n) \to \Lambda(x) \quad \text{for all } \Lambda \in X'.$$

Definition 6.20 (Weak-$*$ topology)

The dual X' carries its own strong and weak topologies, but also a third: the family of seminorms $\{p_x : \Lambda \mapsto |\Lambda(x)| \,:\, x \in X\}$ generates the **weak-$*$ topology** τ_{w*} on X'. Thus $\Lambda_n \overset{*}{\rightharpoonup} \Lambda$ if and only if $\Lambda_n(x) \to \Lambda(x)$ for every $x \in X$: weak-$*$ convergence is pointwise convergence of functionals.

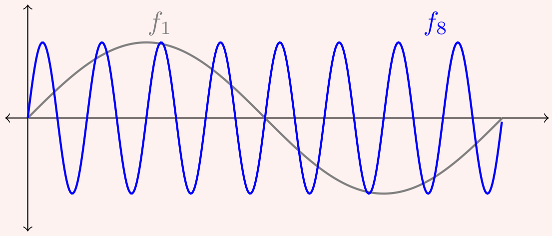
Since $X \subseteq X''$, the weak-$*$ topology on X' is coarser than its weak topology, and the two coincide exactly when X is reflexive. Strong convergence implies weak convergence implies weak-$*$ convergence, and none of the implications reverses.

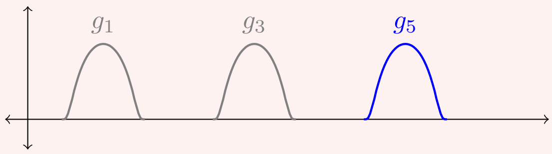
Example 6.21

Two examples of weak convergence to 0 without strong convergence, which between them illustrate the two mechanisms available.

(i) (*Oscillation.*) In $L^2((0,2\pi))$ let $f_n(x) = \sin(nx)$. Then $\|f_n\|_{L^2} = \sqrt{\pi}$ for all n, so $f_n \not\to 0$ strongly. But $\int_0^{2\pi} g(x)\sin(nx)\,\mathrm{d}x \to 0$ for every $g \in L^2$: this is true for g in the dense set of trigonometric polynomials by orthogonality, hence for all g by density and uniform boundedness. So $f_n \rightharpoonup 0$. The mass does not go away; it is redistributed into ever higher frequencies, and every fixed test function averages it out.

(ii) (*Escape to infinity.*) In $L^2(\mathbb{R})$ let $g_n(x) = \phi(x-n)$ for a fixed non-zero bump ϕ. Again $\|g_n\|_{L^2} = \|\phi\|_{L^2}$ is constant, but for $h \in C_c(\mathbb{R})$ the supports eventually separate, so $\int hg_n = 0$ for large n, and density gives $g_n \rightharpoonup 0$. The mass is neither destroyed nor spread out – it simply escapes to infinity.





Example 6.22

Specialising to our spaces: for $1 \le p < \infty$ and $f_i, f \in L^p(\mathbb{R}^n)$, we have $f_i \rightharpoonup f$ if and only if

$$\int_{\mathbb{R}^n} gf_i \longrightarrow \int_{\mathbb{R}^n} gf \quad \text{for all } g \in L^q(\mathbb{R}^n),$$

by Theorem 5.10. When $1 < p < \infty$ the space is reflexive and this is the same as weak-$*$ convergence. For $f_i \in L^\infty(\mathbb{R}^n) = L^1(\mathbb{R}^n)'$, weak-$*$ convergence means testing against all $g \in L^1$, which is strictly weaker than weak convergence, since $(L^\infty)'$ is strictly bigger than L^1.

Remark. Two warnings. First, the weak topology is *not* metrisable on an infinite-dimensional space (though it is metrisable on bounded sets when X' is separable), so one must be careful with sequential arguments; we shall always be in the situation where sequences suffice. Second, weak convergence is strictly weaker than strong convergence, but by Proposition 6.14 a weakly convergent sequence is still bounded, and one always has

$$\|x\| \le \liminf_{n\to\infty} \|x_n\| \quad \text{when } x_n \rightharpoonup x,$$

since $\|x\| = \sup_{\|\Lambda\|\le 1} |\Lambda(x)| = \sup_\Lambda \lim_n |\Lambda(x_n)| \le \liminf_n \|x_n\|$. The norm can drop in the limit – it did in both parts of Example 6.21, from a positive constant to 0 – but it cannot jump up. This inequality, **weak lower semicontinuity of the norm**, is the single most useful property of weak convergence, and everything in Section 7 is built on it.

§6.6 Banach–Alaoglu

Now the pay-off.

Theorem 6.23 (Banach–Alaoglu)

Let X be a normed space and $\overline{B}' = \{\Lambda \in X' : \|\Lambda\|_{X'} \le 1\}$ the closed unit ball of the dual. Then $\overline{B}'$ is compact in the weak-$*$ topology.

The general statement is proved by embedding $\overline{B}'$ into the product $\prod_{x\in X}\{z : |z| \le \|x\|\}$ and applying Tychonoff's theorem, and therefore uses the axiom of choice in an essential way. For our purposes a sequential version on separable spaces is enough, and it can be proved by hand with a diagonal argument.

Theorem 6.24 (Banach–Alaoglu, sequential form)

Let X be a separable Banach space and let $(\Lambda_j)_{j=1}^\infty$ be a sequence in $\overline{B}' \subseteq X'$. Then there is a subsequence (Λ_{j_k}) and $\Lambda \in \overline{B}'$ with $\Lambda_{j_k} \overset{*}{\rightharpoonup} \Lambda$.

Proof. Let $D = \{x_1, x_2, \dots\}$ be a countable dense subset of X.

Diagonal extraction. The sequence $(\Lambda_j(x_1))_j$ is bounded in $\mathbb{F}$, since $|\Lambda_j(x_1)| \le \|x_1\|$. By Bolzano–Weierstrass some subsequence converges, to a limit we call $\Lambda(x_1)$; write $\Lambda_{1,k}$ for this subsequence. Extract from it a further subsequence $\Lambda_{2,k}$ along which $\Lambda_{2,k}(x_2)$ converges, to $\Lambda(x_2)$; and so on. The diagonal sequence $\Lambda_{j,j}$ is eventually a subsequence of each $\Lambda_{l,k}$, so $\Lambda_{j,j}(x) \to \Lambda(x)$ for every $x \in D$, with $|\Lambda(x)| \le \|x\|$.

Extension to X. The map $\Lambda : D \to \mathbb{F}$ is Lipschitz: given $x, y \in D$, choose k so large that $|\Lambda_{k,k}(x) - \Lambda(x)|$ and $|\Lambda_{k,k}(y) - \Lambda(y)|$ are both less than ε; then

$$|\Lambda(x) - \Lambda(y)| \le |\Lambda(x) - \Lambda_{k,k}(x)| + |\Lambda_{k,k}(x-y)| + |\Lambda_{k,k}(y) - \Lambda(y)| \le 2\varepsilon + \|x - y\|.$$

As ε was arbitrary, Λ is 1-Lipschitz on D and so extends uniquely to a continuous map on $X = \overline{D}$.

Linearity. Let $x, y \in X$ and $a \in \mathbb{F}$, and set $z = x + ay$. Choose $x', y', z' \in D$ close to x, y, z respectively with $z' = x' + ay'$ approximately; then

$$\begin{aligned}|\Lambda(z) - \Lambda(x) - a\Lambda(y)| \le{} & |\Lambda(z) - \Lambda(z')| + |\Lambda(x) - \Lambda(x')| + |a||\Lambda(y) - \Lambda(y')| \\ & + |\Lambda(z') - \Lambda_{k,k}(z')| + |\Lambda(x') - \Lambda_{k,k}(x')| + |a||\Lambda(y') - \Lambda_{k,k}(y')| \\ & + |\Lambda_{k,k}(z' - x' - ay')|,\end{aligned}$$

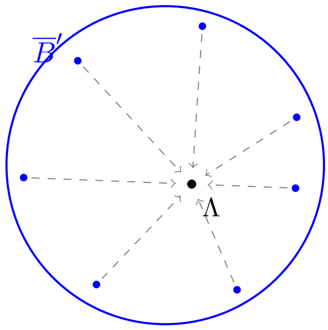
and each line is small: the first by continuity of Λ and the choice of x', y', z', the second by taking k large, the third because $\Lambda_{k,k}$ is linear of norm at most 1 and $z' - x' - ay'$ is small. So Λ is linear, and being 1-Lipschitz, $\Lambda \in \overline{B}'$.

Convergence. Finally, for arbitrary $x \in X$ pick $x' \in D$ with $\|x - x'\|$ small; then

$$|\Lambda_{j,j}(x) - \Lambda(x)| \le |\Lambda_{j,j}(x - x')| + |\Lambda_{j,j}(x') - \Lambda(x')| + |\Lambda(x' - x)|,$$

where the outer terms are at most $\|x - x'\|$ each and the middle tends to 0. Hence $\Lambda_{j,j} \xrightharpoonup{*} \Lambda$. ☐

The theorem is worth pausing over. The closed unit ball of X' is never norm-compact in infinite dimensions, and yet it is always weak-$*$ compact. Nothing has been gained for free: what we get in exchange for compactness is that the limit is only a weak-$*$ limit, so that the convergence carries very little information. The whole art of the subject is to use structure – convexity, an equation, an estimate – to upgrade the weak limit afterwards.



$\Lambda_j \xrightharpoonup{*} \Lambda$, but $\|\Lambda_j - \Lambda\|_{X'} \not\to 0$

Corollary 6.25

Let $1 < p \le \infty$ and let (f_j) be a sequence in $L^p(\mathbb{R}^n)$ with $\|f_j\|_{L^p} \le K$ for all j. Then there is a subsequence (f_{j_k}) and $f \in L^p(\mathbb{R}^n)$ with $\|f\|_{L^p} \le K$ and

$$\int_{\mathbb{R}^n} f_{j_k} g \, \mathrm{d}x \longrightarrow \int_{\mathbb{R}^n} f g \, \mathrm{d}x \quad \text{for all } g \in L^q(\mathbb{R}^n).$$

Proof. $L^p(\mathbb{R}^n) = L^q(\mathbb{R}^n)'$ by Theorem 5.10, and $L^q(\mathbb{R}^n)$ is separable since $q < \infty$ (Proposition 3.14). Apply Theorem 6.24 to $K^{-1} f_j$. ☐

Corollary 6.26

In a reflexive Banach space X (in particular in any Hilbert space, or in L^p for $1 < p < \infty$), every bounded sequence has a weakly convergent subsequence.

Proof. Apply Theorem 6.24 to the sequence viewed in $X'' = (X')'$, using that weak-$*$ and weak convergence coincide on a reflexive space. (For the separability hypothesis one restricts to the closed span of the sequence, which is a separable reflexive space.) ☐

Notice that $p = 1$ is excluded, and this is not an oversight. In $L^1(\mathbb{R})$ the sequence $f_n = n\mathbb{1}_{[0,1/n]}$ is bounded, but it has no weakly convergent subsequence: the only candidate limit is the Dirac mass at 0, which is not a function. Failures of weak compactness in L^1 are always of this type – concentration – and controlling them is the subject of the

Dunford–Pettis theorem, which we shall not need.

§7 Convexity and the Direct Method

We now set out, in the abstract, the method by which we shall later prove that partial differential equations have solutions. The idea goes back to Hilbert and is disarmingly simple: to find a minimiser of a functional S, take a sequence u_n along which $S[u_n]$ tends to the infimum, extract a convergent subsequence, and hope that the limit is a minimiser. Every step of this can fail, and the interest is entirely in the conditions which make it work.

§7.1 Lower semicontinuity

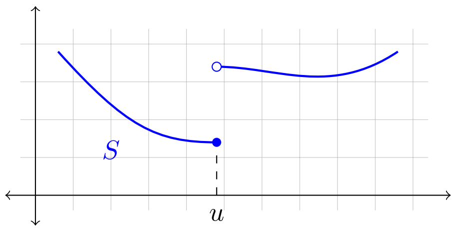
Definition 7.1

Let X be a topological space. A functional $S : X \to \mathbb{R} \cup \{+\infty\}$ is **(sequentially) lower semicontinuous** if

$$u_n \to u \implies S[u] \leq \liminf_{n\to\infty} S[u_n].$$

It is **coercive** if every sublevel set $\{u : S[u] \leq K\}$ is sequentially relatively compact – that is, if $S[u_n] \leq K$ for all n then (u_n) has a convergent subsequence.

Continuity implies lower semicontinuity, but not conversely. The point of the weaker hypothesis is that the value of S is allowed to *drop* in a limit but not to jump up; and dropping is harmless when we are minimising.



Theorem 7.2 (The direct method)

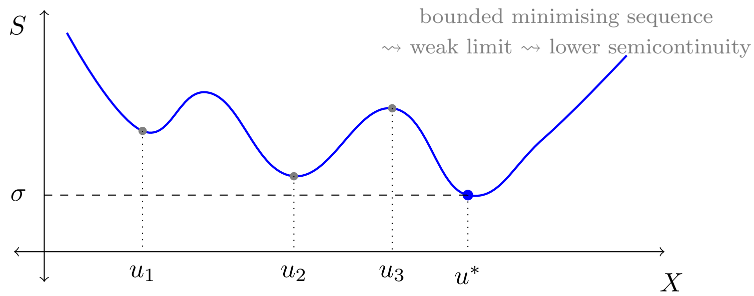
Let X be a topological space and $S : X \to \mathbb{R}$ coercive and lower semicontinuous. Then S attains its infimum on X.

Proof. Let $\sigma = \inf\{S[u] : u \in X\} \in [-\infty, \infty)$ and choose a **minimising sequence** (u_n) with $S[u_n] \to \sigma$. In particular $(S[u_n])$ is bounded above, say by K, so coercivity provides a subsequence $u_{n_j} \to u^*$ for some $u^* \in X$. By lower semicontinuity,

$$S[u^*] \leq \liminf_{j\to\infty} S[u_{n_j}] = \sigma.$$

As $S[u^*] \geq \sigma$ by definition of the infimum, we get $S[u^*] = \sigma$; in particular $\sigma > -\infty$. $\square$

That is the entire method, and as stated it is nearly vacuous. All the work lies in verifying the two hypotheses, and here we run into a fundamental tension. Coercivity asks for compactness, which wants a *weak* topology; lower semicontinuity asks for the functional to behave well under limits, which wants a *strong* topology. Any given functional will typically satisfy one and fail the other. Resolving this is the content of the rest of the section, and the resolution is convexity.



§7.2 Coercivity in the weak topology

From now on let X be a reflexive separable Banach space. On such a space, coercivity in the weak topology is nothing more than an a priori bound.

Lemma 7.3

Let X be reflexive and suppose $S : X \to \mathbb{R}$ has the property that $S[u] \leq K$ implies $\|u\| \leq \widetilde{K}$ for some $\widetilde{K}$ depending only on K. Then S is coercive for the weak topology of X.

Proof. Immediate from Corollary 6.26: a bounded sequence has a weakly convergent subsequence. □

So on a reflexive space, coercivity is free – it is just the statement that sublevel sets of S are bounded, which for a concrete functional is a routine estimate. The difficulty has been entirely displaced onto lower semicontinuity, which we must now verify for the weak topology. Since weak convergence is a much weaker hypothesis on (u_n), weak lower semicontinuity is a much stronger conclusion, and it is simply false in general. (Consider $S[u] = -\|u\|^2$ on a Hilbert space and $u_n = e_n$: then $u_n \rightharpoonup 0$ but $S[u_n] = -1 < 0 = S[0]$.)

§7.3 Convexity saves the day

Lemma 7.4 (Mazur)

Let X be a Banach space and $U \subseteq X$ convex and strongly closed. Then U is weakly closed.

Proof. Let $x \in U^c$. Applying Theorem 6.8(b) to the compact convex set $\{x\}$ and the closed convex set U gives $\Lambda \in X'$ and $\gamma_1 < \gamma_2$ with $\operatorname{Re}\Lambda(x) < \gamma_1 < \gamma_2 < \operatorname{Re}\Lambda(y)$ for all $y \in U$. The set $\{z : \operatorname{Re}\Lambda(z) < \gamma_1\}$ is weakly open (it is a preimage of an open set under a weakly continuous map), contains x, and is disjoint from U. So U^c is weakly open. □

The reader should appreciate how much this says. A strongly closed set need not be weakly closed – the unit *sphere* of a Hilbert space is strongly closed but its weak closure is the whole closed unit ball, since $e_n \rightharpoonup 0$. Convexity rules out exactly this kind of behaviour.

Theorem 7.5

Let X be a Banach space and $S : X \to \mathbb{R}$ convex and lower semicontinuous for the strong topology. Then S is lower semicontinuous for the weak topology.

Proof. For each $K \in \mathbb{R}$ the sublevel set $E_K = \{u : S[u] \leq K\}$ is convex (by convexity of S) and strongly closed (by strong lower semicontinuity), hence weakly closed by Lemma 7.4.

Now suppose $u_n \rightharpoonup u^*$, and set $\ell = \liminf_n S[u_n]$; we may assume $\ell < \infty$. Passing to a subsequence we may assume $S[u_{n_j}] \to \ell$. Given $\varepsilon > 0$, there is J with $S[u_{n_j}] \leq \ell+\varepsilon$ for $j \geq J$, i.e. $u_{n_j} \in E_{\ell+\varepsilon}$ for $j \geq J$. Since $E_{\ell+\varepsilon}$ is weakly closed and $u_{n_j} \rightharpoonup u^*$, we get $u^* \in E_{\ell+\varepsilon}$, that is $S[u^*] \leq \ell + \varepsilon$. As $\varepsilon > 0$ was arbitrary, $S[u^*] \leq \ell$. □

Putting the pieces together we obtain the theorem we shall use.

Corollary 7.6

Let X be a reflexive Banach space and $S : X \to \mathbb{R}$ convex, strongly lower semicontinuous, and with bounded sublevel sets. Then S attains its infimum on X.

Proof. Combine Lemma 7.3, Theorem 7.5 and Theorem 7.2, all applied to the weak topology. (Strictly, Theorem 7.2 was stated for sequential compactness and sequential lower semicontinuity, which is exactly what the three results provide.) □

Example 7.7

The norm itself is convex and continuous, so $u \mapsto \|u\|$ is weakly lower semicontinuous: this recovers the inequality $\|u^*\| \leq \liminf \|u_n\|$ for $u_n \rightharpoonup u^*$ that we saw directly earlier. More usefully, so is $u \mapsto \|u\|^2 - 2\operatorname{Re}\Lambda(u)$ for a fixed $\Lambda \in X'$, and this functional has bounded sublevel sets since

$$\|u\|^2 - 2\operatorname{Re}\Lambda(u) \geq \|u\|^2 - 2\|\Lambda\|\,\|u\| \to \infty$$

as $\|u\| \to \infty$. So it attains its infimum. In a Hilbert space this is a long-winded proof of the Riesz representation theorem; on a Sobolev space it will be a proof that the Dirichlet problem has a solution.

Remark. It is worth isolating the shape of the argument, because it recurs in every application. To minimise S over X:

(i) show S is bounded below and take a minimising sequence (u_n);

(ii) derive an a priori bound $\|u_n\| \leq K$ from $S[u_n] \leq \sigma+1$ – this is a *coercivity* estimate and is where the structure of S enters;

(iii) extract $u_{n_j} \rightharpoonup u^*$ by reflexivity;

(iv) show $S[u^*] \leq \liminf S[u_{n_j}]$ using convexity;

(v) conclude that u^* is a minimiser, and derive its Euler–Lagrange equation by differentiating $t \mapsto S[u^* + tv]$ at $t = 0$.

Step (v) is what connects minimisation to differential equations, and we carry it out in

Section **??**.